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Notice

This expansion pack requires an M1C0001 Pybit car, a micro:bit v2, and 3 AA batteries. Please ensure you have these basic components.

It is not sold together with the M1C0001 Pybit robot car—you will need to prepare it separately. It is compatible with standard LEGO® building blocks. If you have the same building blocks listed in our tutorial, you can follow along with the instructions to build.

If you do not have exactly the same building blocks as listed, but you have sufficient experience to build functionally similar building block expansion models, you may also proceed—functional code is provided for your use.

If you wish to purchase this expansion pack, please contact our sales representatives via our official website.

Safety Warning

Choking Hazard: The building blocks required for this tutorial contain small parts. Keep them away from children under 3 years old to prevent accidental swallowing.

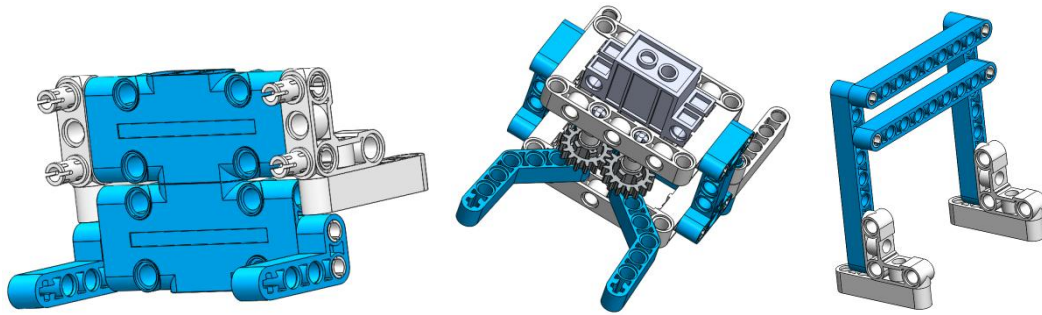
Electrical Safety: Before starting the building blocks assembly, ensure the Micro:bit and Pybit are powered off.

2

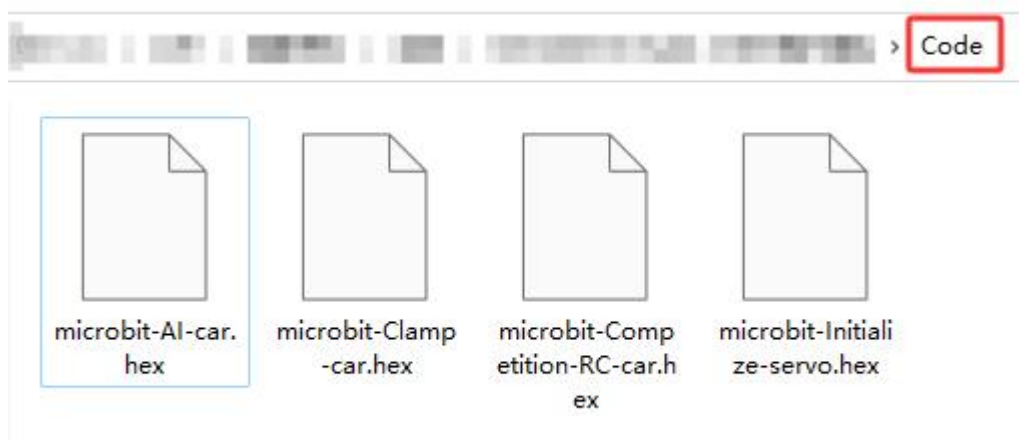
Expansion Pack Introduction

To meet the demand of Pybit users for smart interactive experiences, we have created its dedicated building block expansion pack. This enables the Pybit to perform simple pushing of objects (such as a small ball), gripping small objects with a mechanical claw, and recognizing road signs with an AI camera to move according to the directional instructions in the images.

In summary, the expansion pack can be assembled into a simple scoop, a mechanical claw, and an AI camera. Please note: All components must be purchased separately.

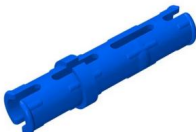


The code used in this tutorial is stored in the following path of the tutorial package: xx/xx/Code.



3

Parts List

Lego Components		
2780 Friction Pin 26PCS	6558 Friction Pin 10 PCS	55615 Right-Angle Pin 6PCS
		

43857 2-Hole Beam 6PCS	32316 5-Hole Beam 6PCS	40490 9-Hole Beam 5PCS
		
32140 L-Beam 2PCS	71709 Panel 2PCS	64179 Circular Beam 2 PCS
		
14720 H-Beam 2PCS	32348 7-Hole Beam 2PCS	3713 Axle Bush 4 pcs
		
32073 Axle 2PCS	94925 16-Tooth Gear 2 PCS	3648 24-Tooth Gear 1PCS
		
Additional Components		
270° Servo Motor 1PCS	Small board 1PCS	KOI2 AI Camera 1PCS
		

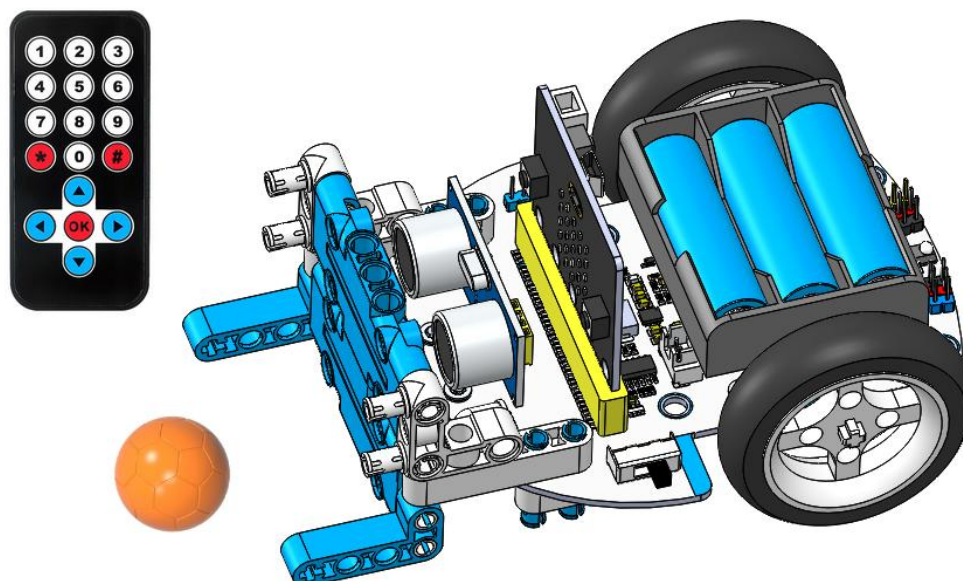
Note: The physical parts listed above must be purchased separately!

4

Competition Car

4.1 Goal

Build a Pybit competition car equipped with a scoop.



4.2 Component Required

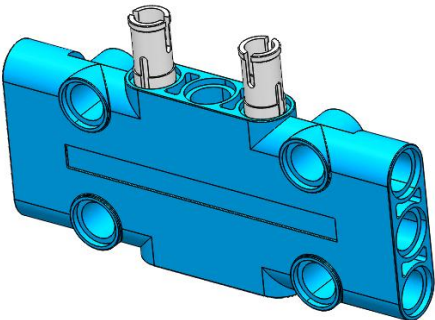
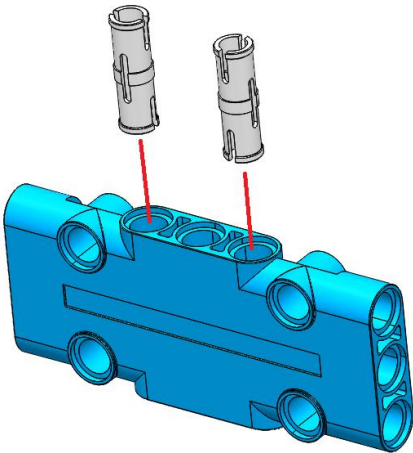
Pybit car with micro:bit v2	IR remote control
Pybit expansion pack	

4.3 Assembly Steps

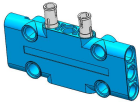
Construct a scoop using standard building blocks and attach it to the Pybit.

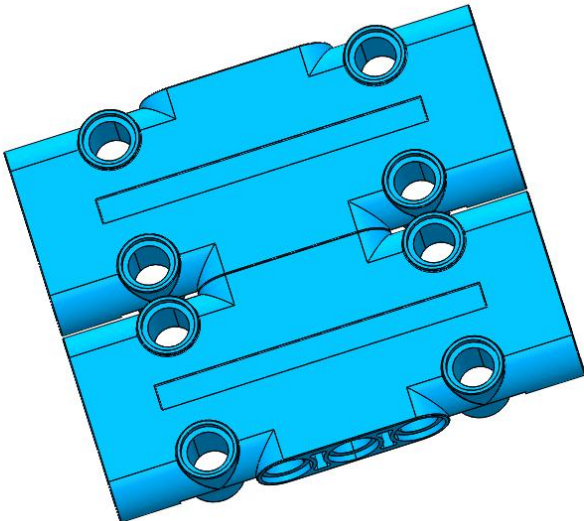
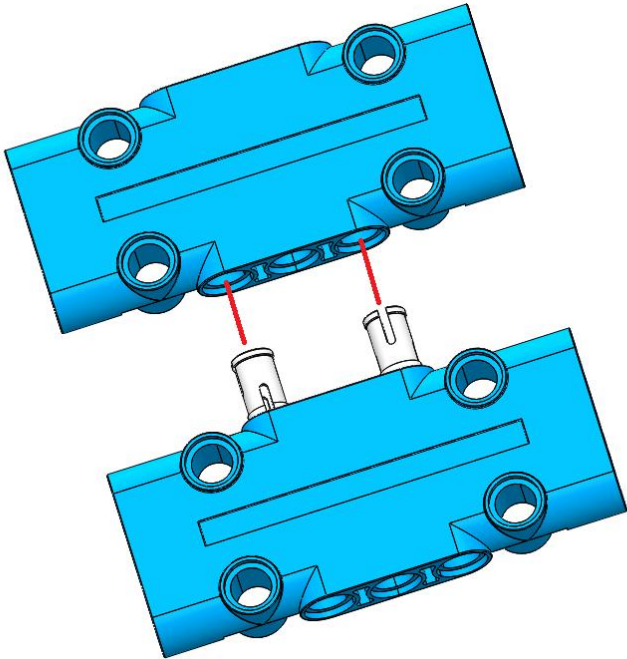
Step 1: Insert pins into the center of the baffle.

2780 Friction Pin 2PCS	71709 Panel 1PCS
	

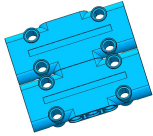


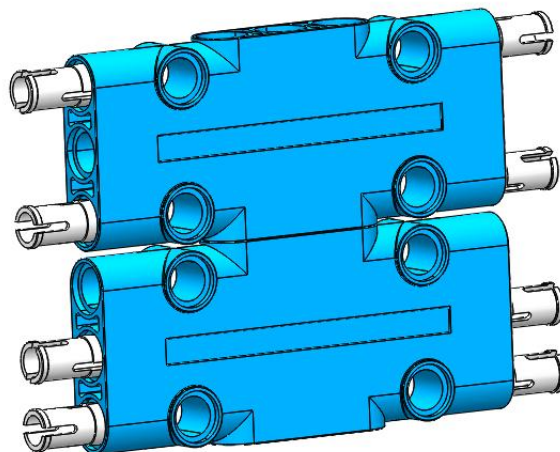
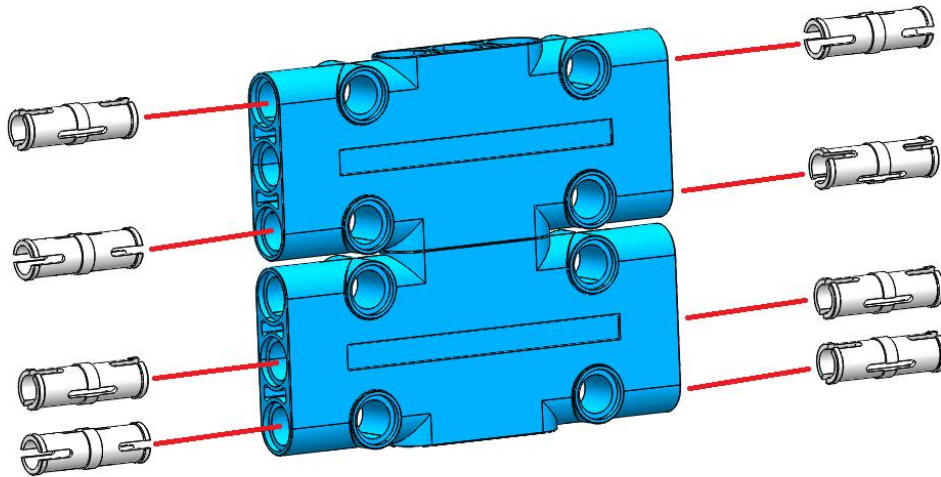
Step 2: Assemble the two baffles.

Component from Step 1	71709 Panel 1PCS
	

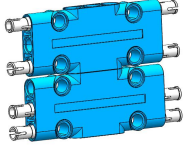


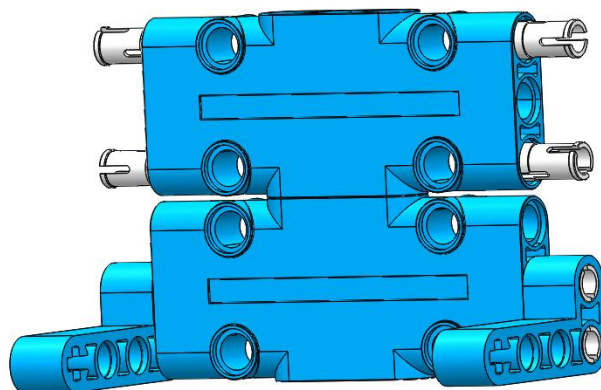
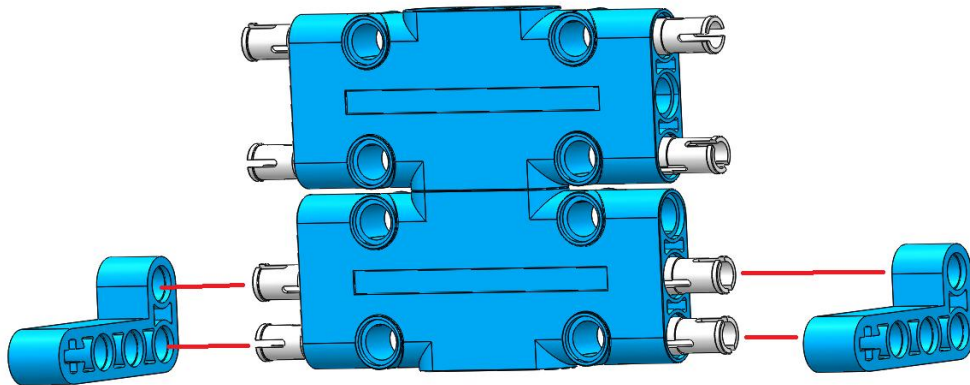
Step 3: Install the side pins on the baffle

Component from Step 2	2780 Friction Pin 8PCS
	



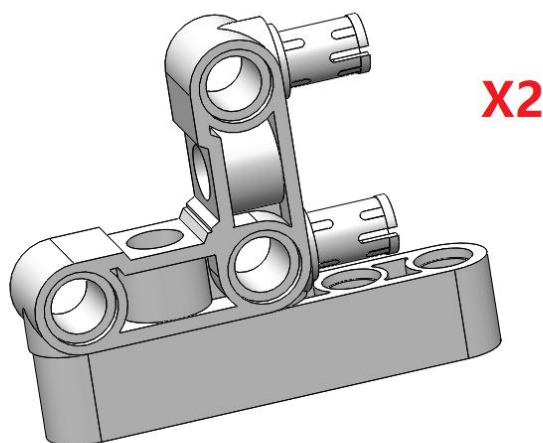
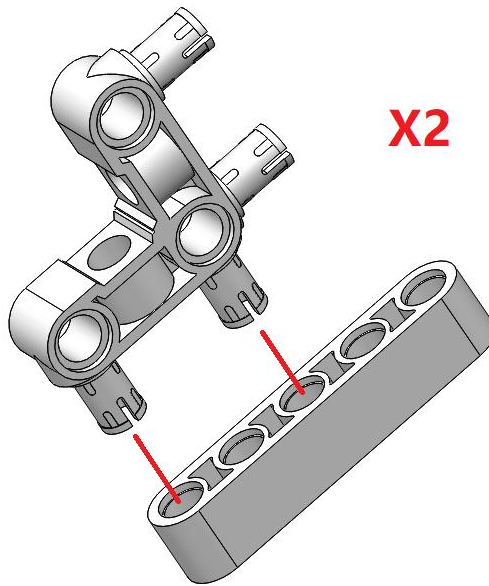
Step 4: Build the bucket teeth

Component from Step 3	32140 L-Beam 2PCS
	

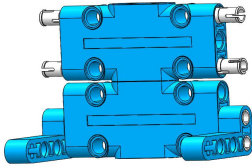
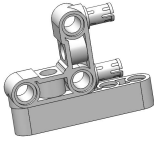


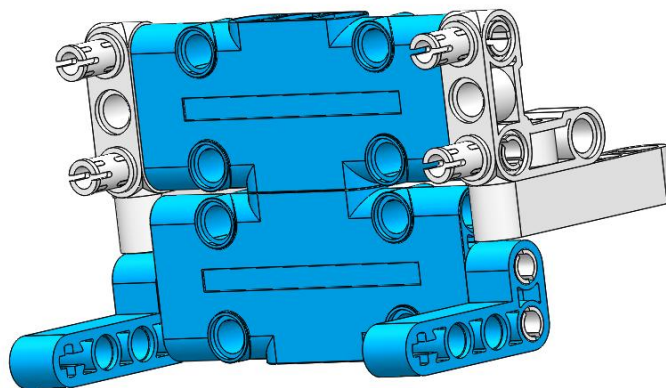
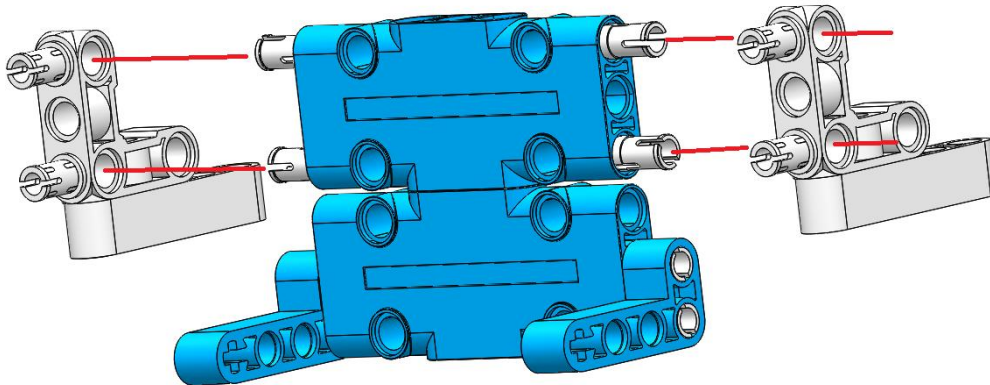
Step 5: Construct the bucket arm

55615 Right-Angle Pin 2PCS	32316 5-Hole Beam 2PCS
	

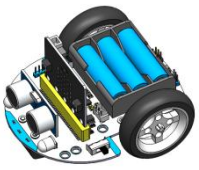


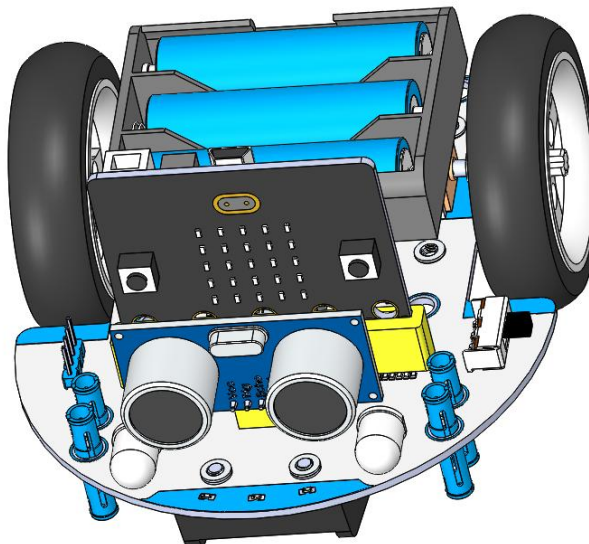
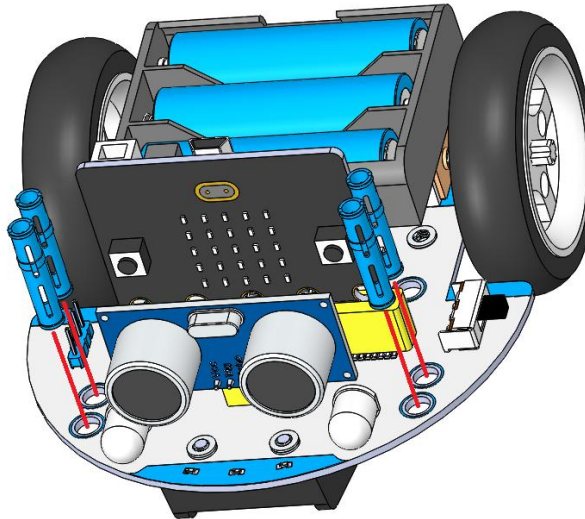
Step 6: Attach the bucket arm to the scoop

Component from Step 4	Component from Step 5 2PCS
	

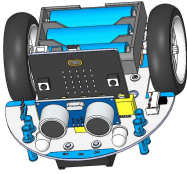


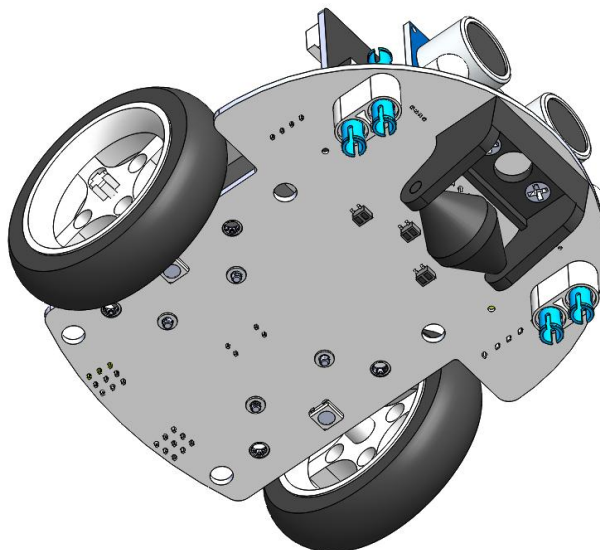
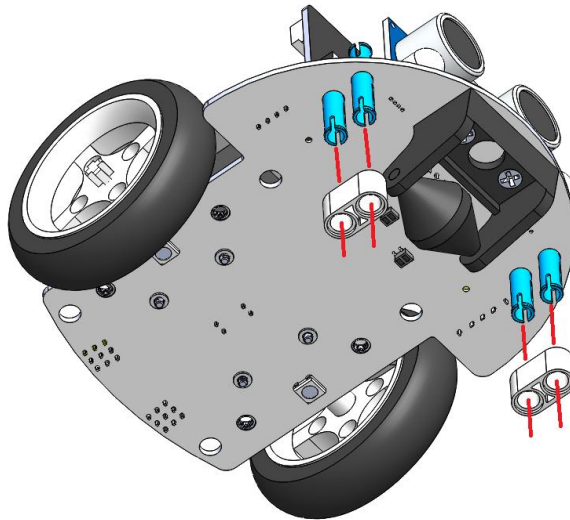
Step 7: Mount the pins onto the Pybit

Pybit car with micro:bit v2	6558 Friction Pin 4 PCS
	

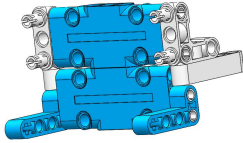
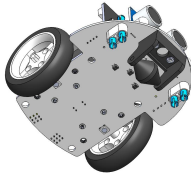


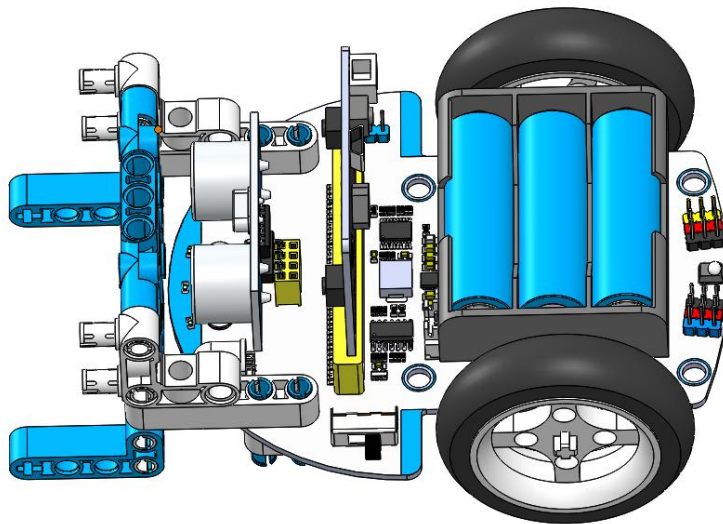
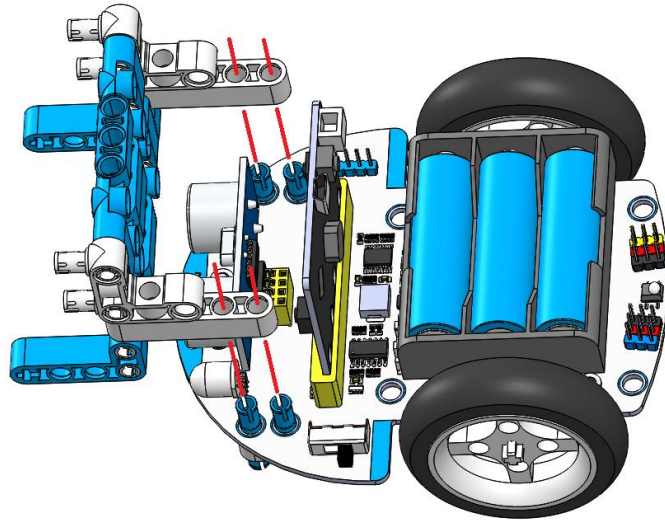
Step 8: Secure the pins on the Pybit

Component from Step 7	43857 2-Hole Beam 2PCS
	



Step 9: Attach the scoop to the Pybit

Component from Step 6	Component from Step 8
	

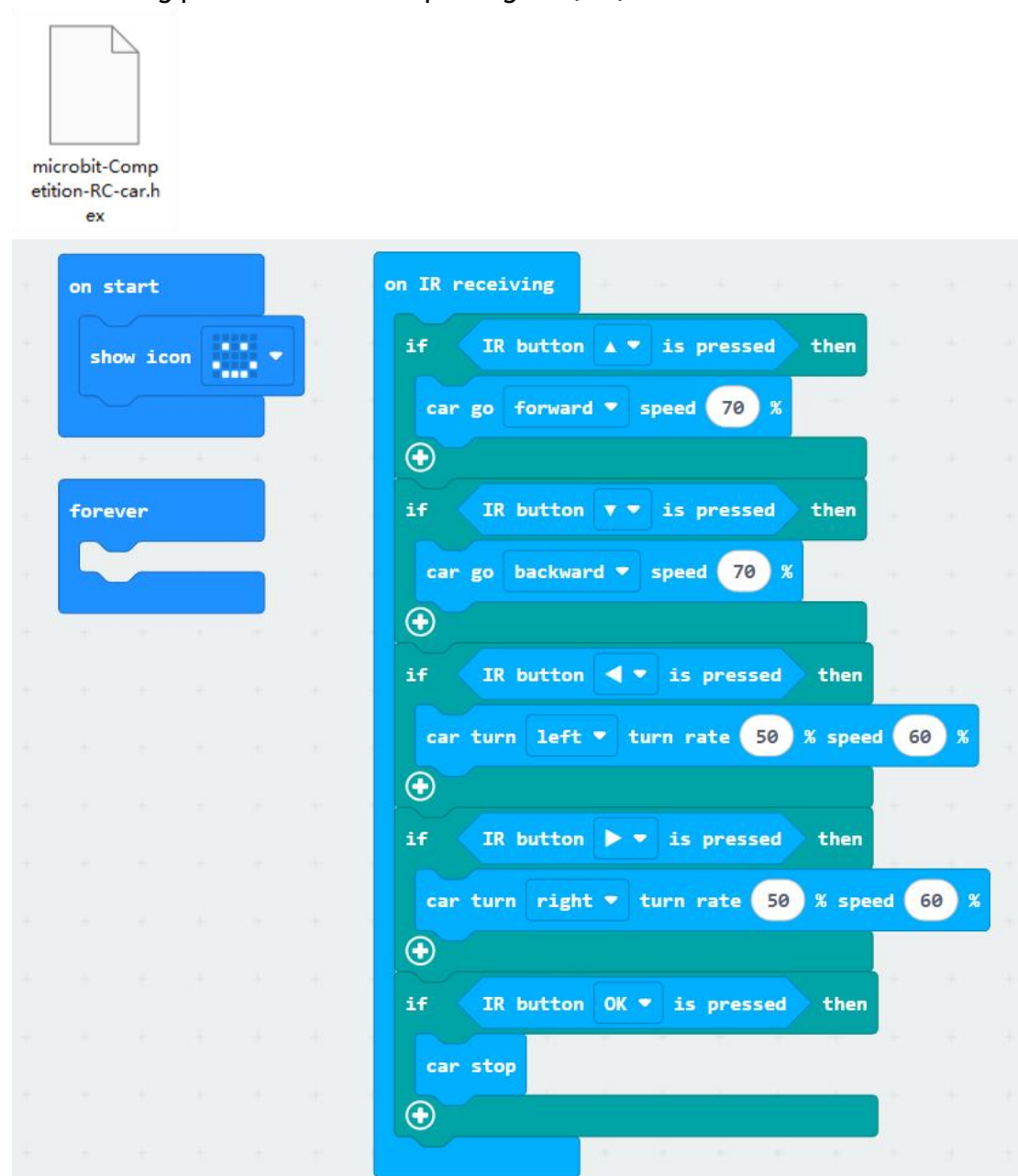


4.4 MakeCode Editor

Visit <https://makecode.microbit.org/>

4.5 Example Code

You can directly upload the code we provided to the Micro:bit on the Pybit car to control the competition car. The example code for the competition car is stored in the following path of the tutorial package: xx/xx/Code.

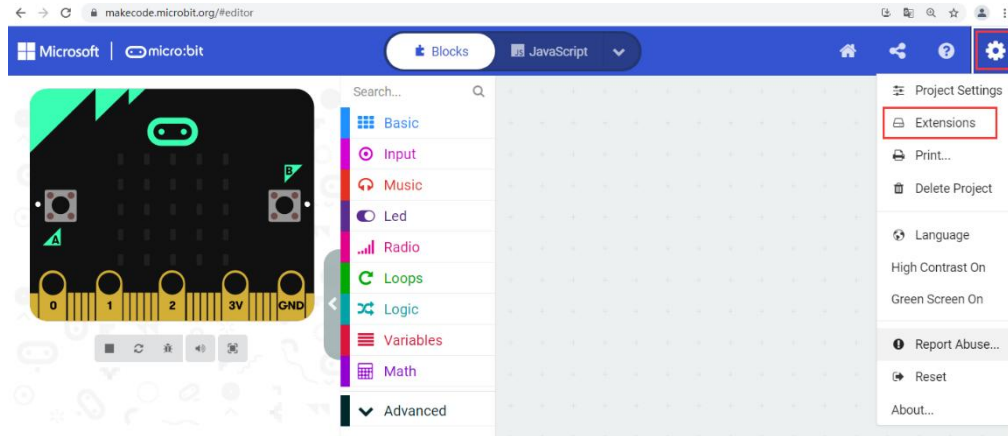


The screenshot displays the MakeCode editor interface. On the left, the 'on start' block contains a 'show icon' block with a car icon. Below it is a 'forever' loop block. On the right, the 'on IR receiving' block contains five 'if' statements for controlling the car's movement based on IR button presses:

- if IR button ▲ is pressed then
car go forward speed 70 %
- if IR button ▼ is pressed then
car go backward speed 70 %
- if IR button ◀ is pressed then
car turn left turn rate 50 % speed 60 %
- if IR button ▶ is pressed then
car turn right turn rate 50 % speed 60 %
- if IR button OK is pressed then
car stop

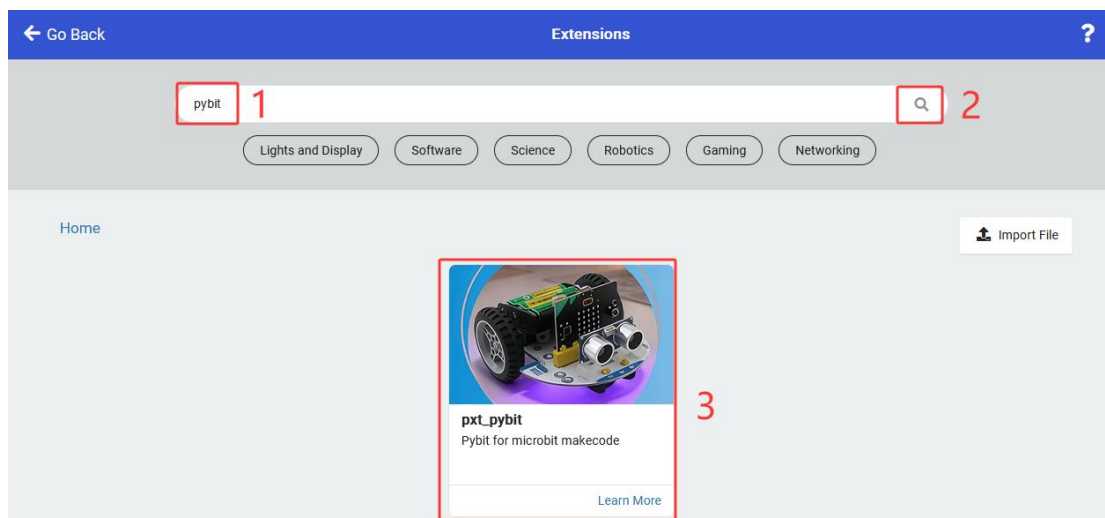
4.6 Extension Library

If you prefer to drag and drop code blocks to create the sample program yourself, you will need to add an extension library specifically developed for Pybit.

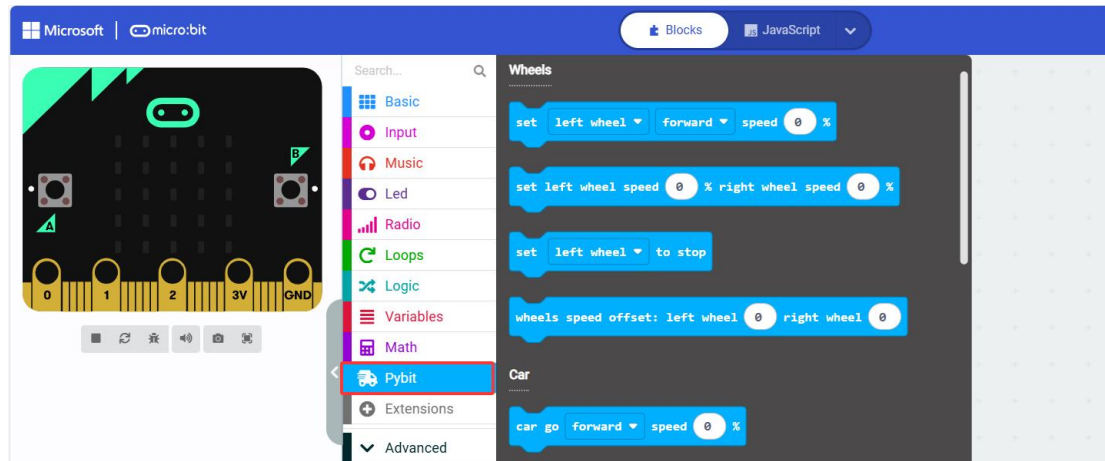


Copy "**pybit**" into the search box on the extensions page, then click the search button on the right.

Click the extension named **Pybit** from the search results.



After a few seconds, the page will redirect to the main MakeCode interface, where you will see the newly added Pybit extension in the toolbox list.



4.7 Result



: Press this button once – the car moves forward



: Press this button once – the car moves backward



: Press this button once – the car turns left



: Press this button once – the car turns right



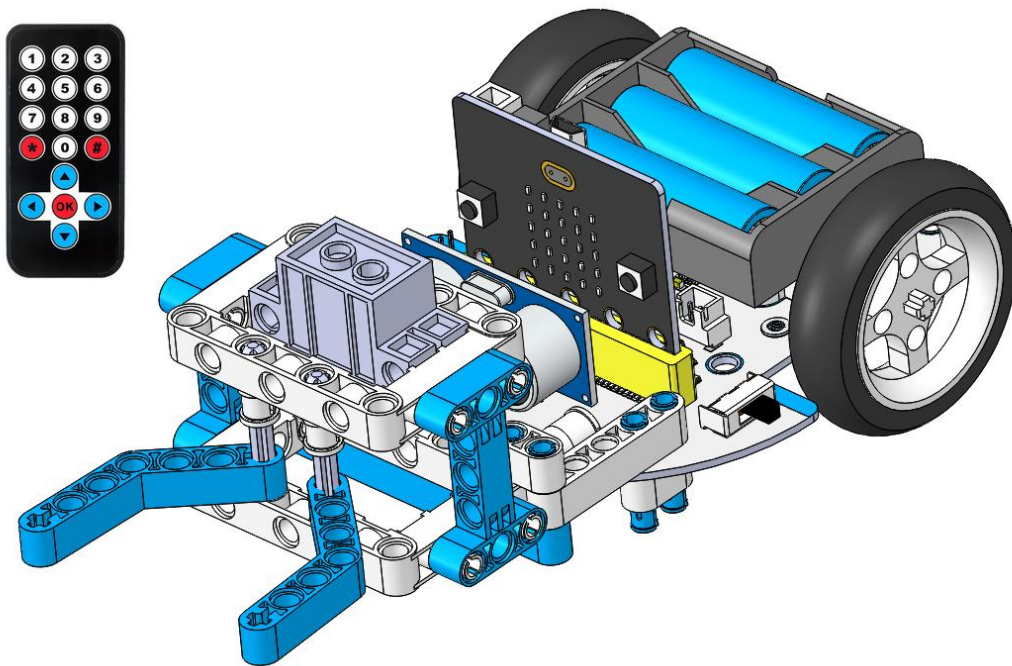
: Press this button once – the car stops.

5

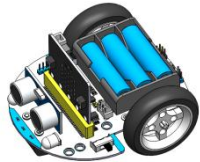
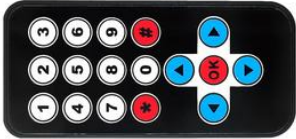
Gripper Car

5.1 Goal

Build a Pybit car equipped with a gripper.



5.2 Component Required

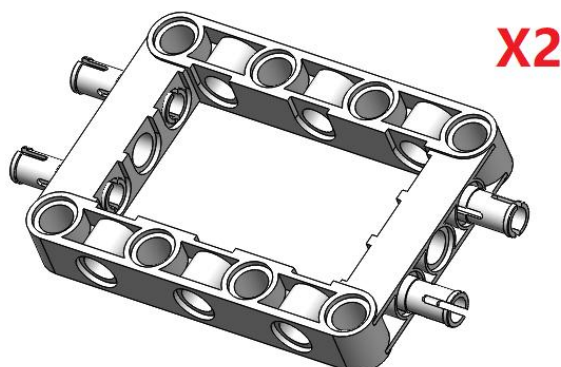
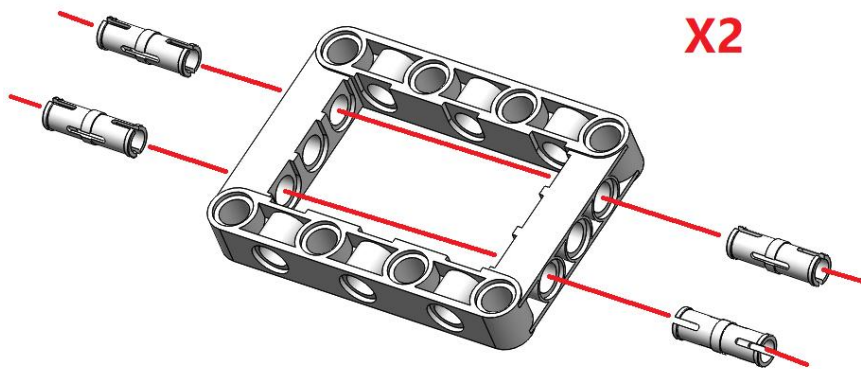
Pybit car with micro:bit v2	IR Remote Control
	
Pybit expansion pack	
	

5.3 Assembly Steps

Construct a gripper using standard building blocks and attach it to the Pybit

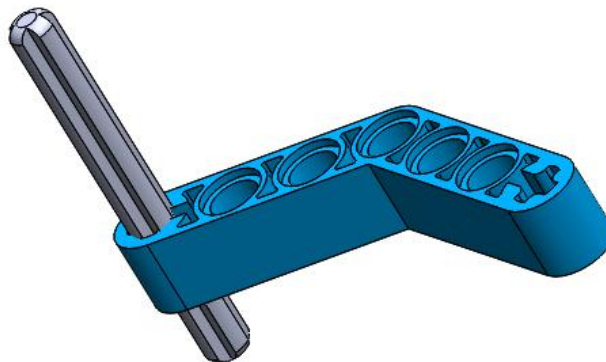
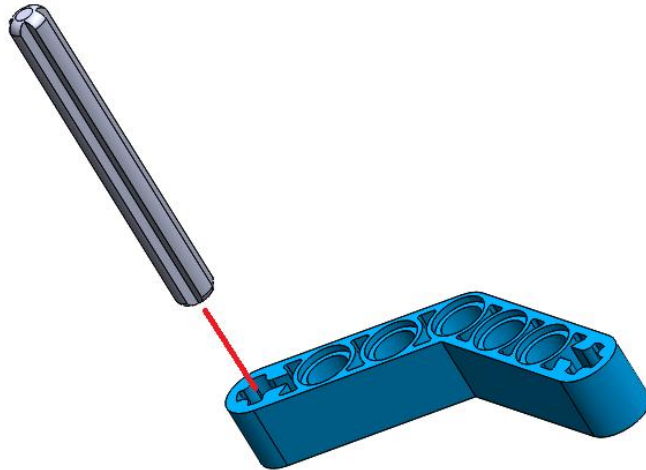
Step 1: Insert pins into the ring beam

2780 Friction Pin 8PCS	64179 Circular Beam 2PCS
	

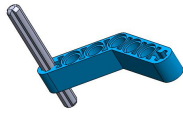


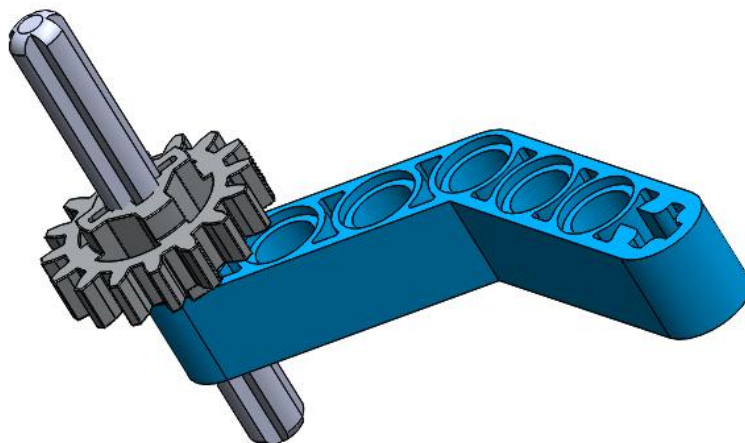
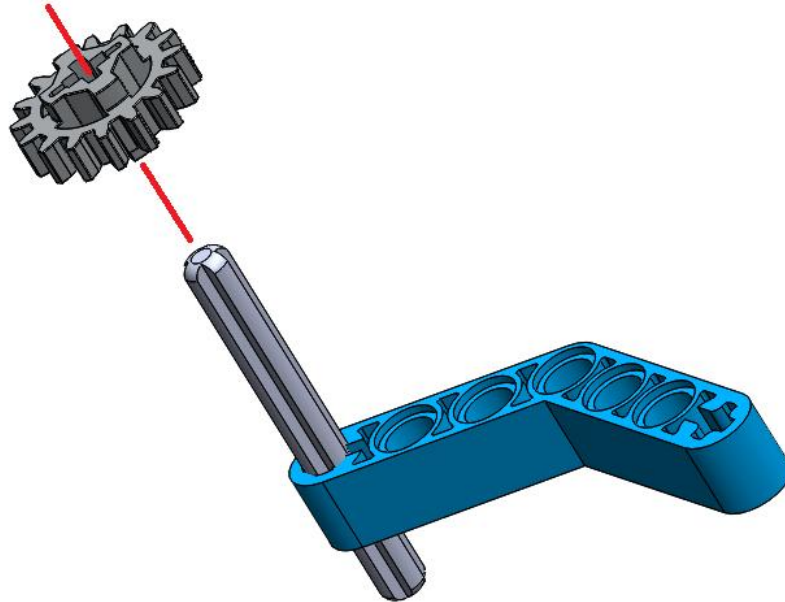
Step 2: Build the left claw axle

32348 7-Hole Beam 1PCS	32073 Axle 1PCS
	

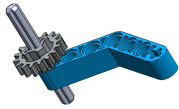


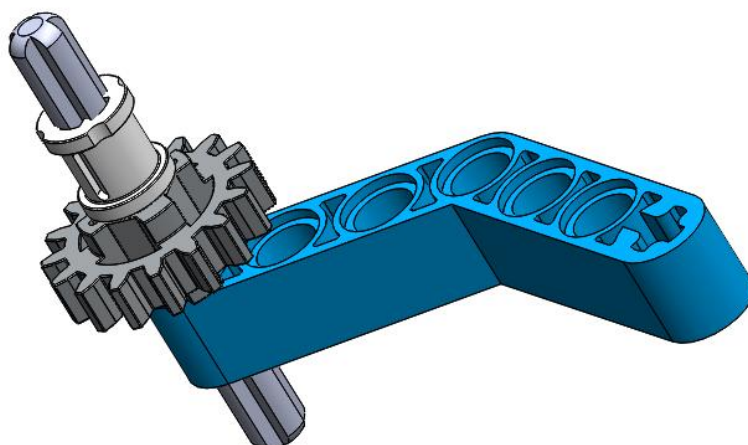
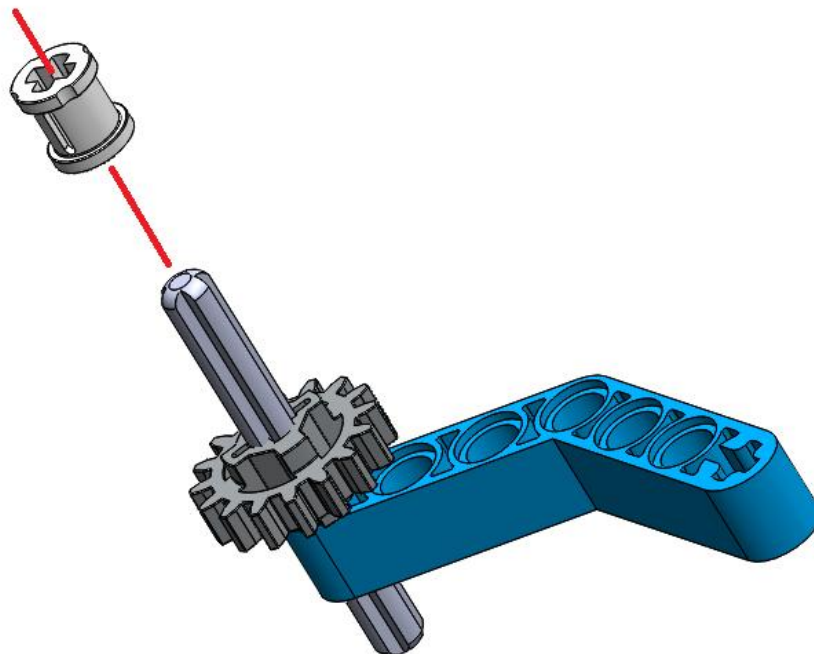
Step 3: Assemble the left claw gear

Component from Step 2	94925 16-Tooth Gear 1PCS
	



Step 4: Attach the left claw axle sleeve

Component from Step 3	3713 Axle Bush 1PCS
	



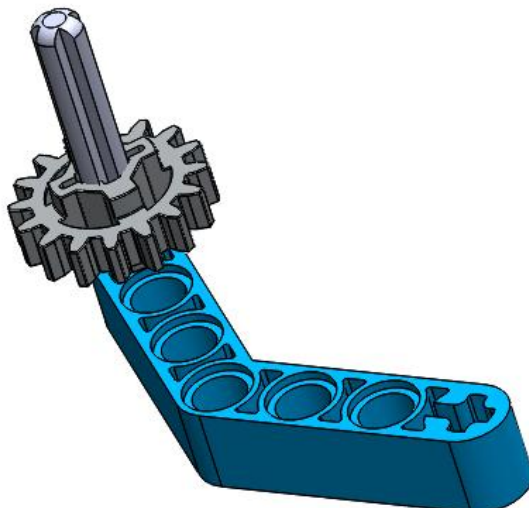
Step 5: Build the right claw axle

32348 7-Hole Beam 1PCS	32073 Axle 1PCS
	



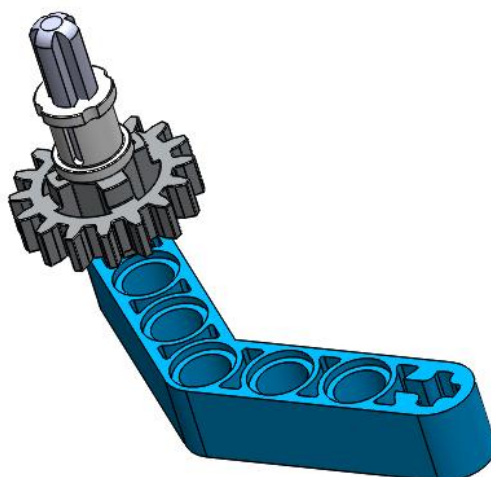
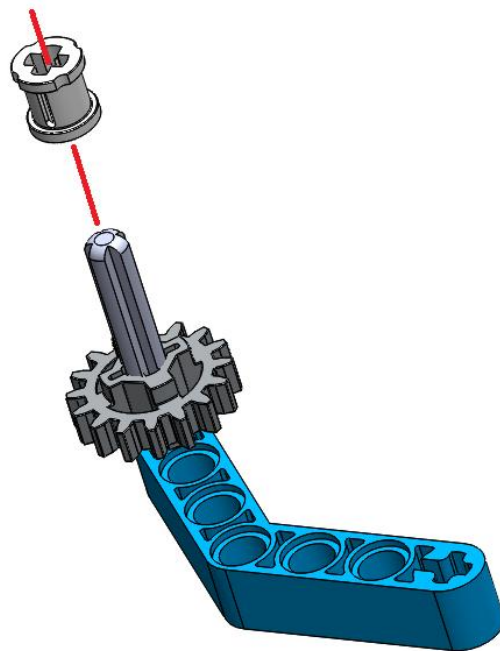
Step 6: Assemble the right claw gear

Component from Step 5	94925 16-Tooth Gear 1PCS
	

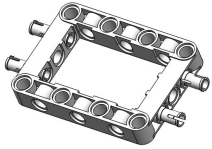
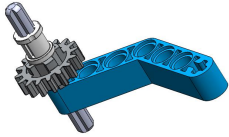
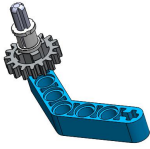


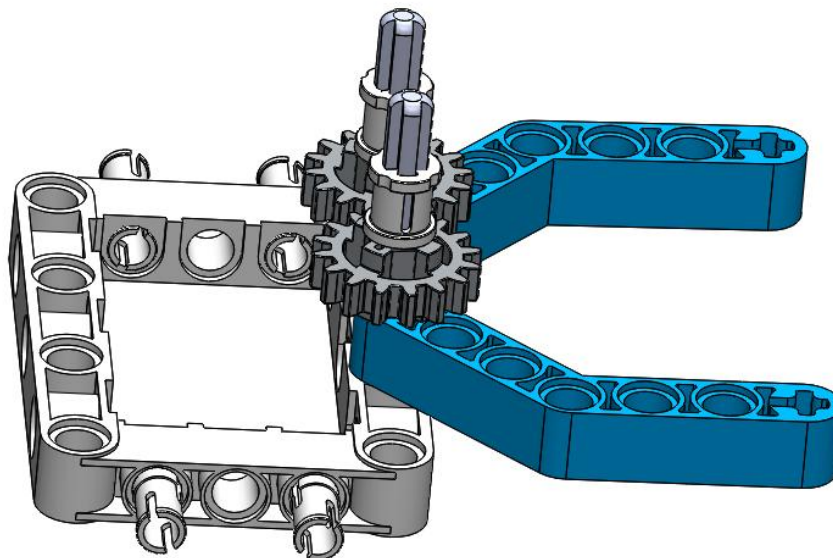
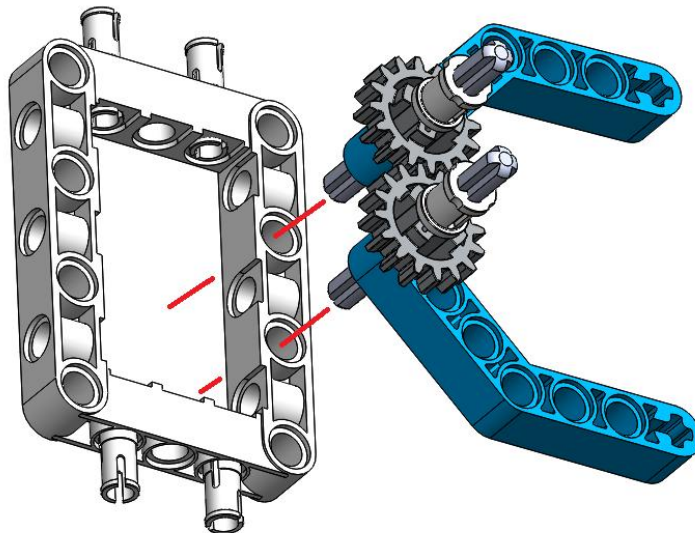
Step 7: Attach the right claw axle sleeve

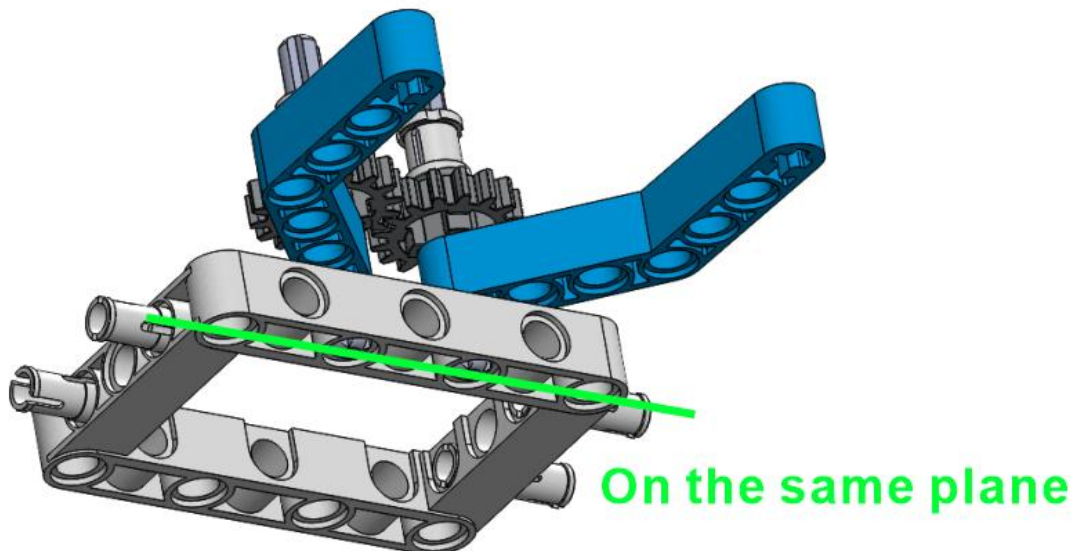
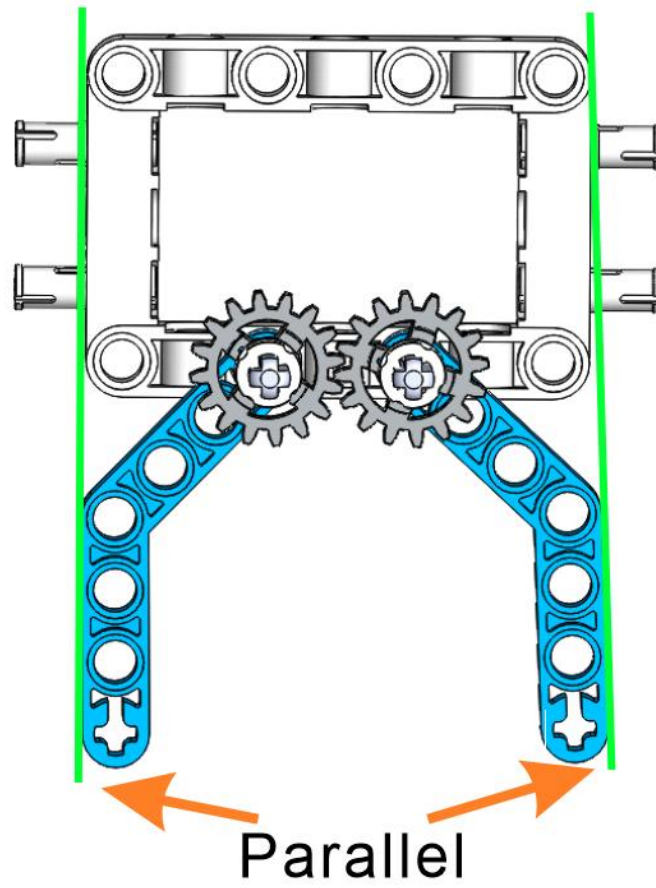
Component from Step 6	3713 Axle Bush 1PCS
	



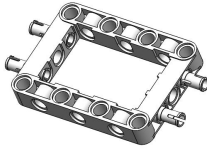
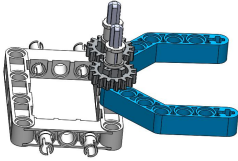
Step 8: Construct the left and right claws

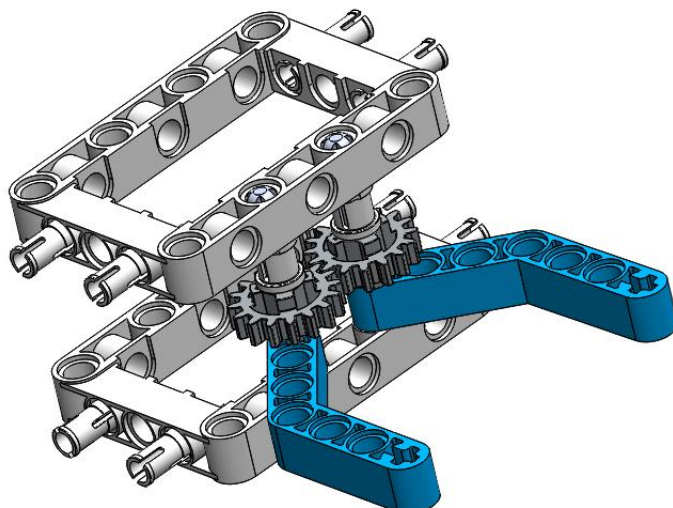
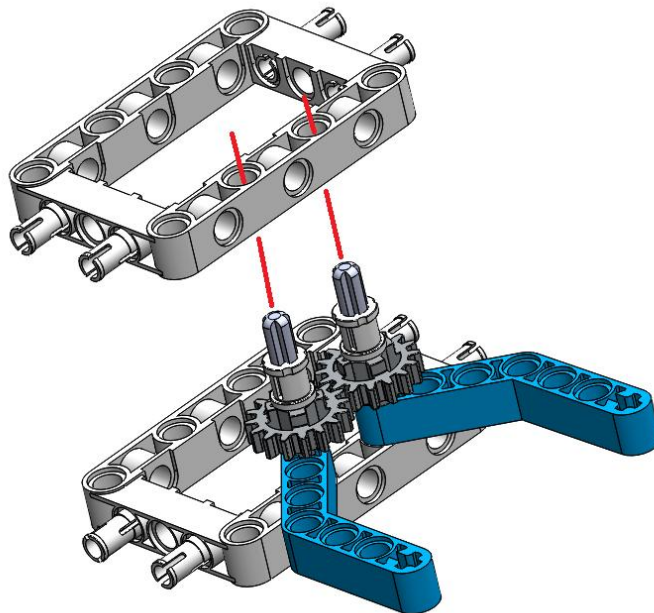
Component from Step 1 1PCS	Component from Step 4	Component from Step 7
		



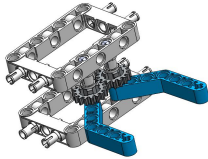


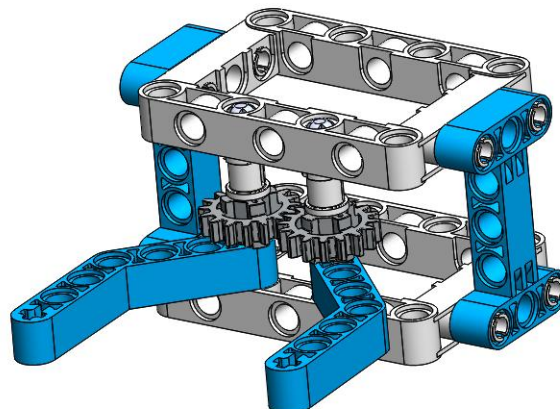
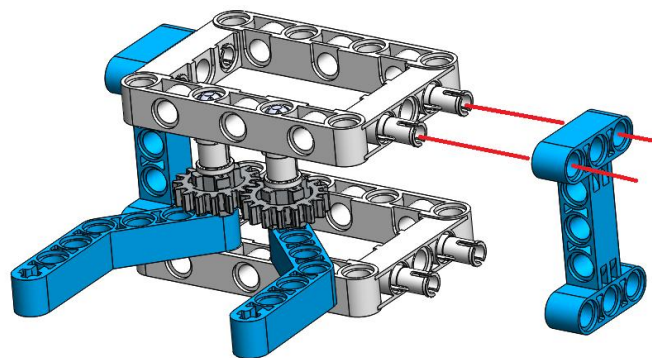
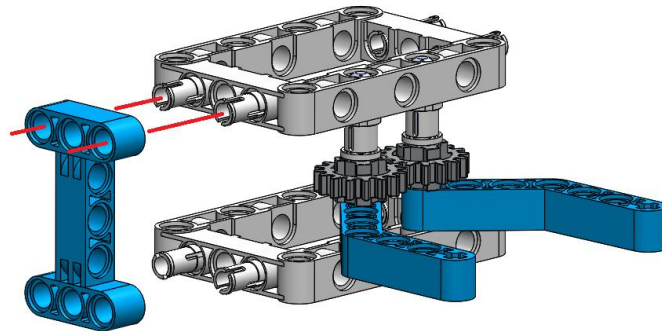
Step 9: Build the claw ring beam

Component from Step 1 1PCS	Component from Step 8
	

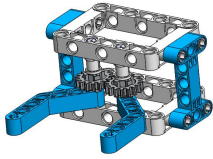


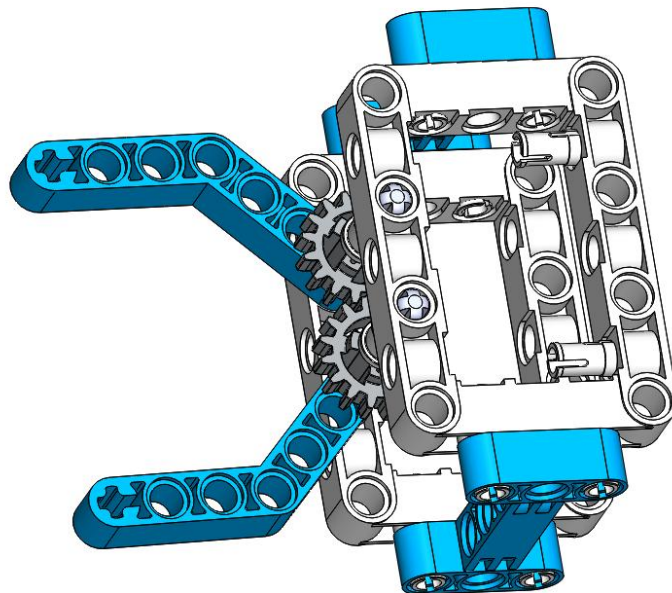
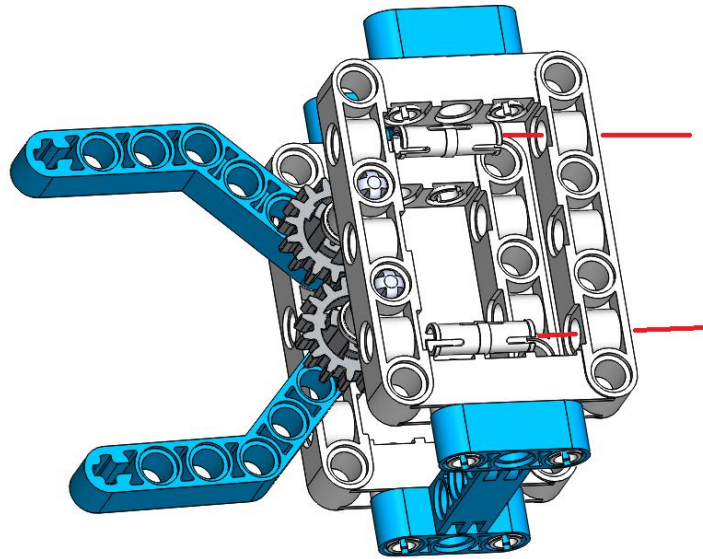
Step 10: Secure the ring beam

Component from Step 9	14720 H-Beam 2PCS
	

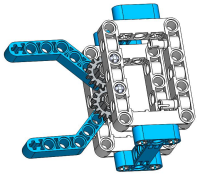


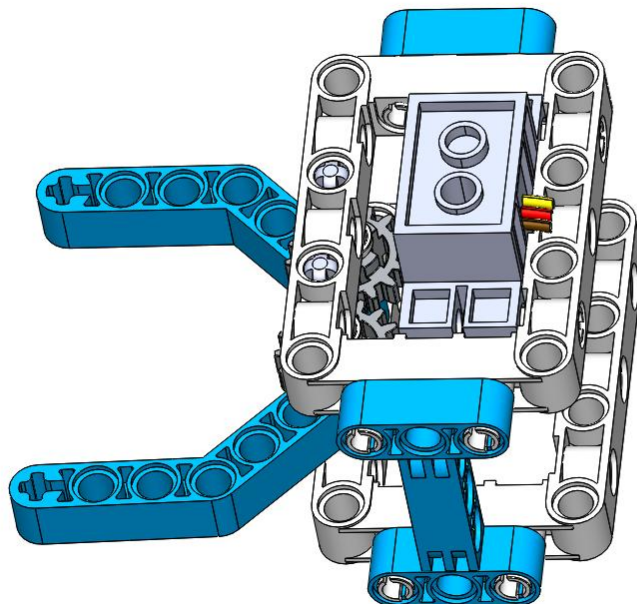
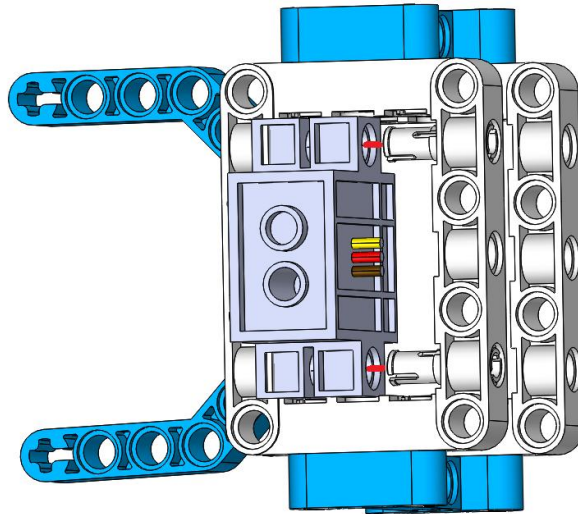
Step 11: Install servo mounting pins

Component from Step 10	2780 Friction Pin 2PCS
	

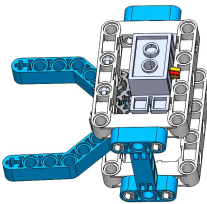


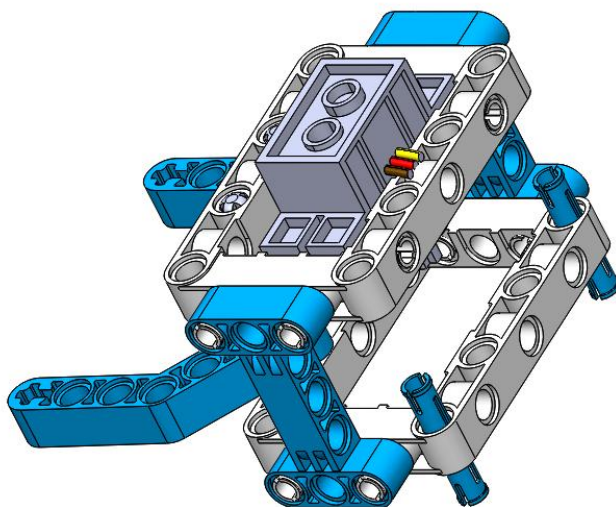
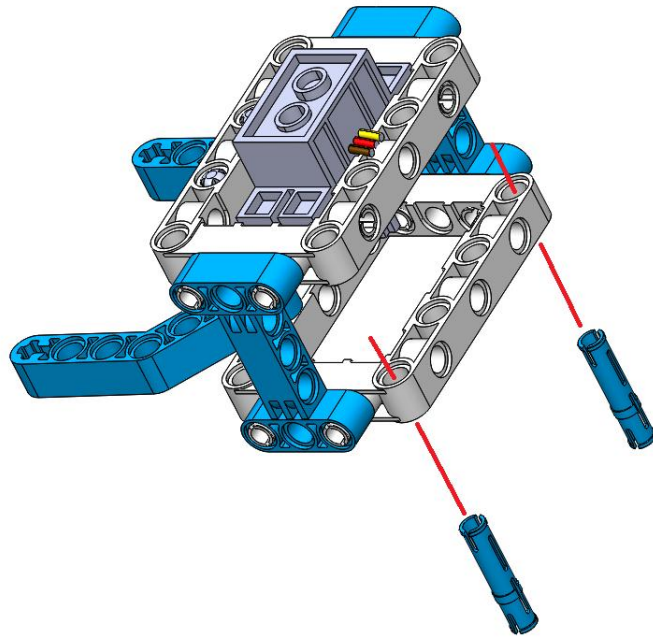
Step 12: Mount the servo motor

Component from Step 11	270° Servo Motor 1PCS
	

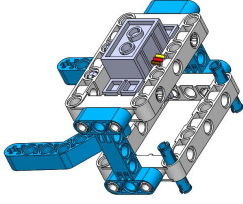


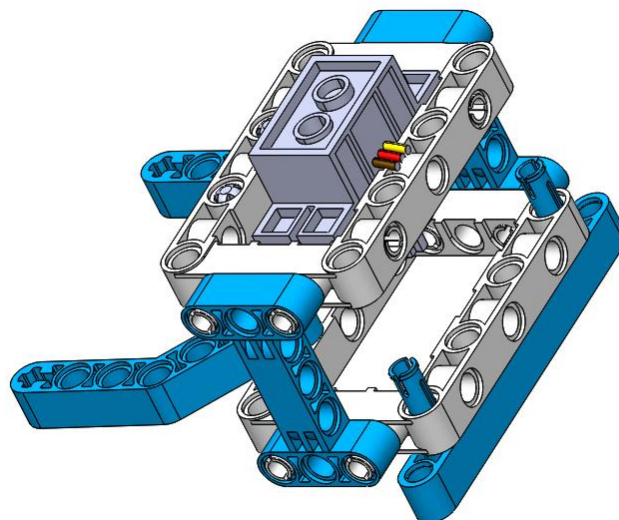
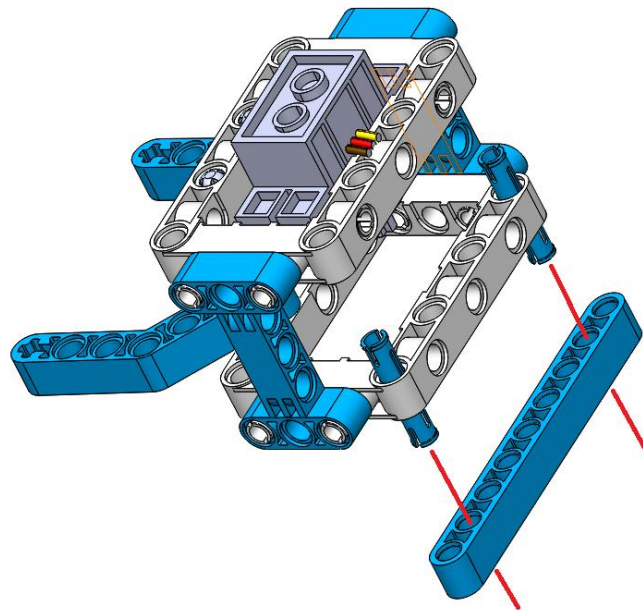
Step 13: Install base beam pins

Component from Step 12	6558 Friction Pin 2 PCS
	

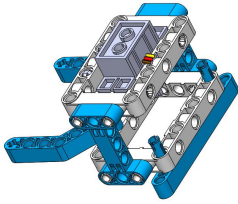


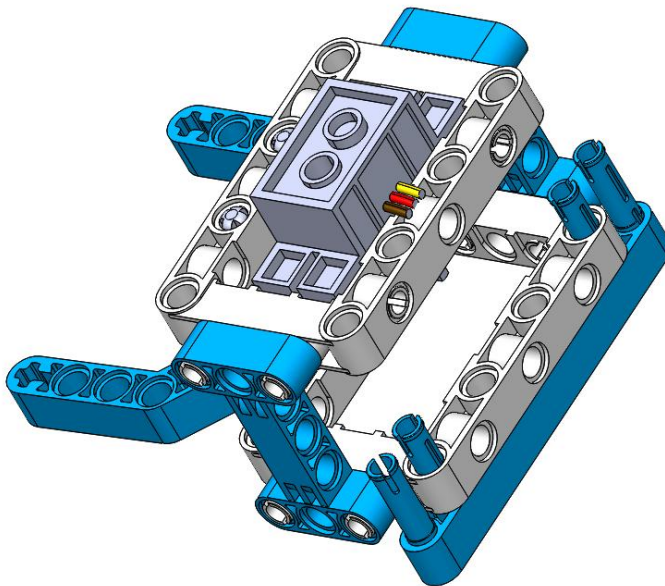
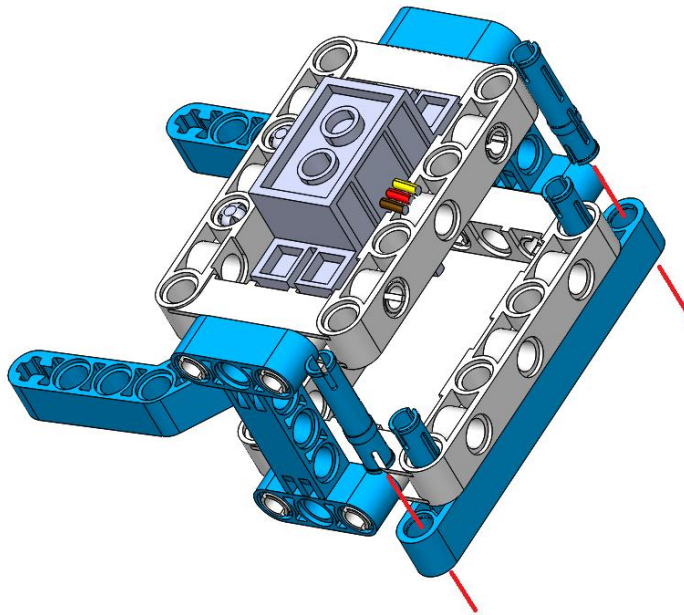
Step 14: Secure the base beam

Component from Step 13	40490 9-Hole Beam 1PCS
	

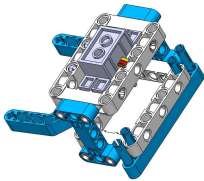


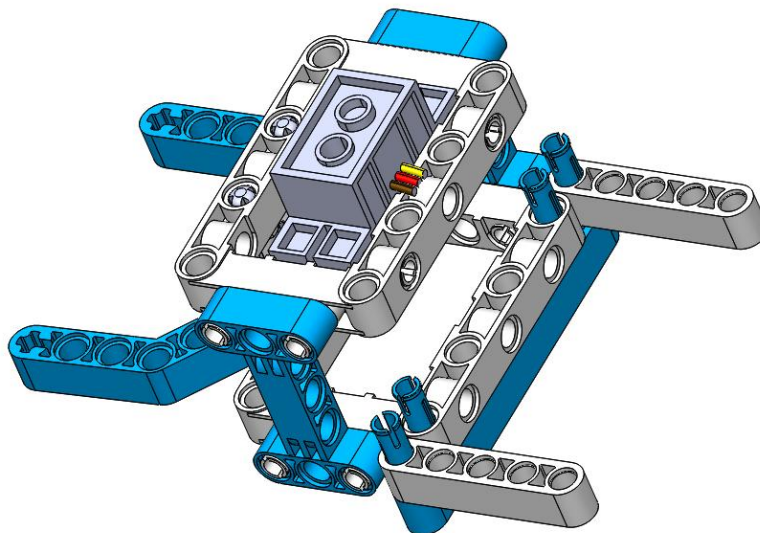
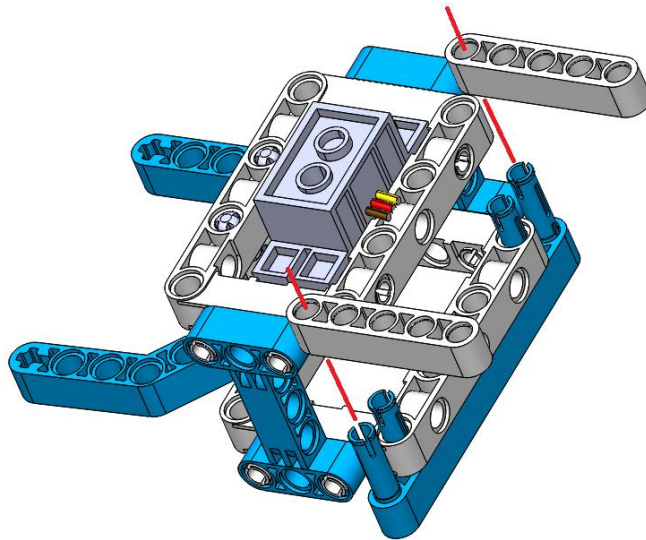
Step 15: Install arm beam pins

Component from Step 14	6558 Friction Pin 2 PCS
	

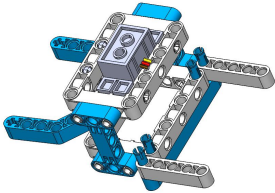


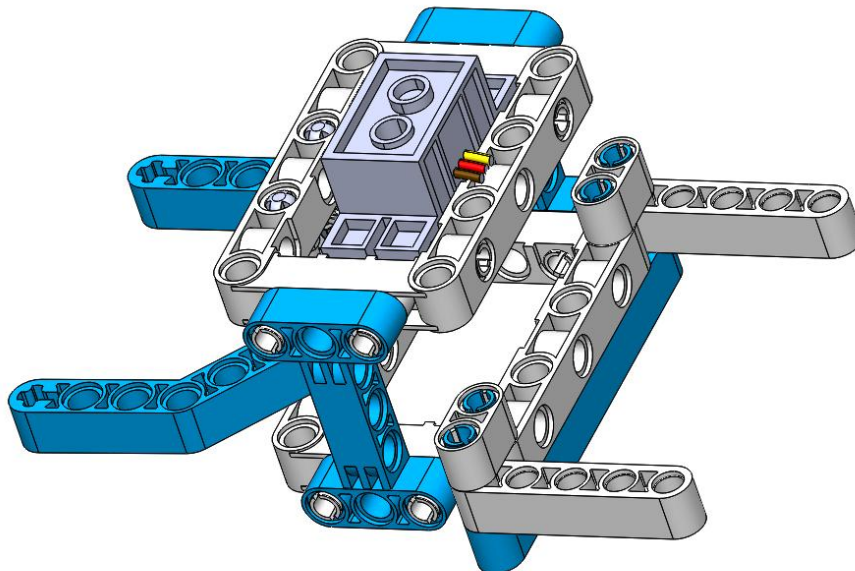
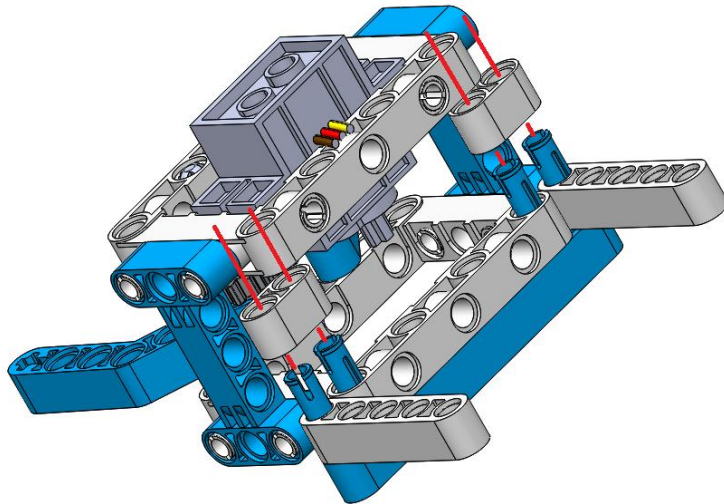
Step 16: Construct the arm beam

Component from Step 15	32316 5-Hole Beam 2PCS
	

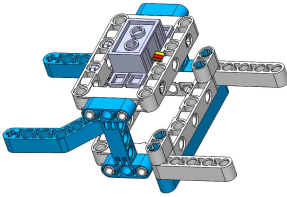
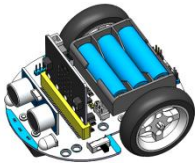


Step 17: Secure the arm beam

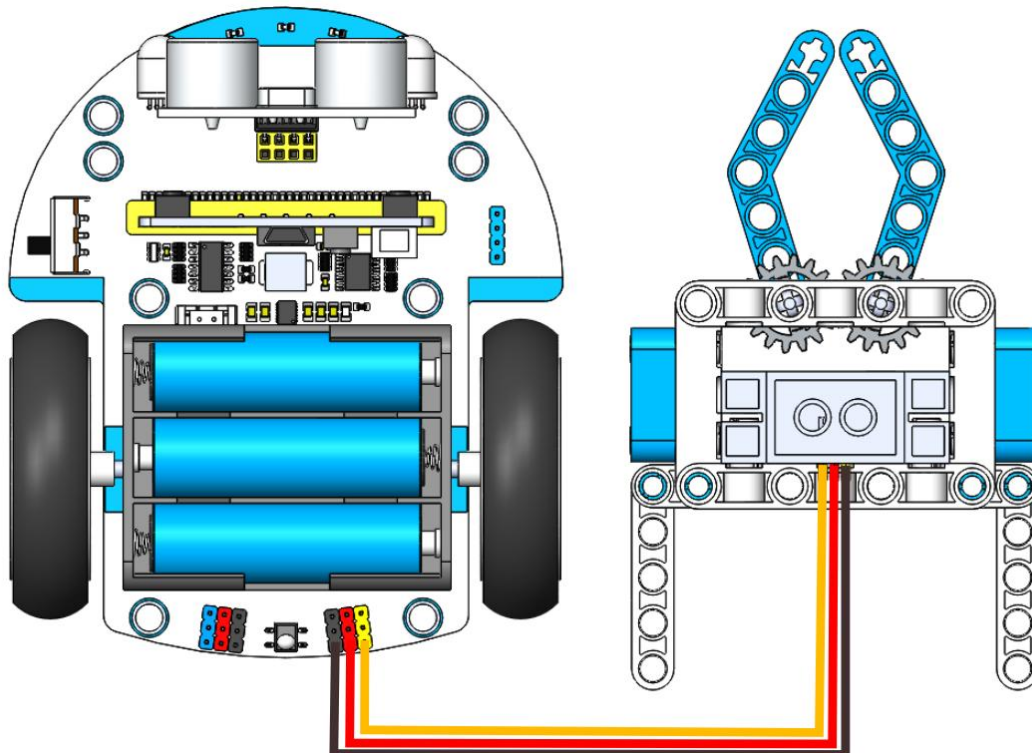
Component from Step 16	43857 2-Hole Beam 2PCS
	



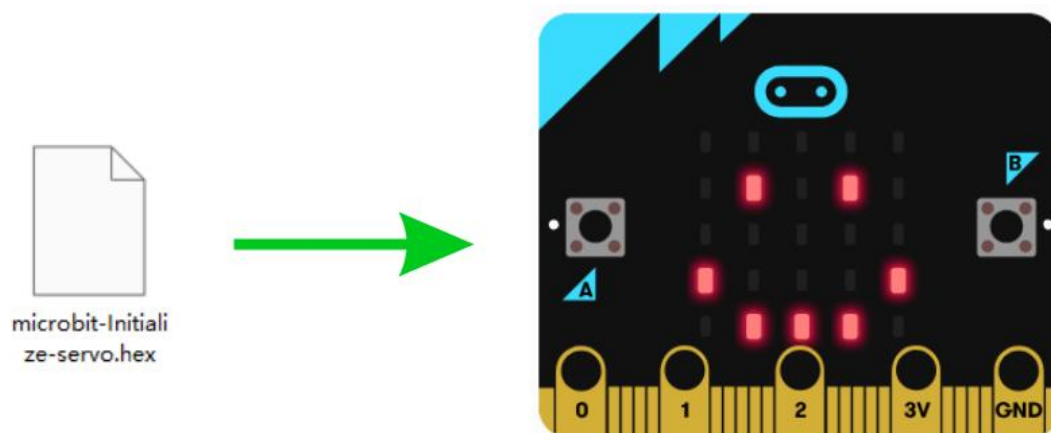
Step 18: Initialize the servo angle (Important!)

Component from Step 17	Pybit car with micro:bit v2
	

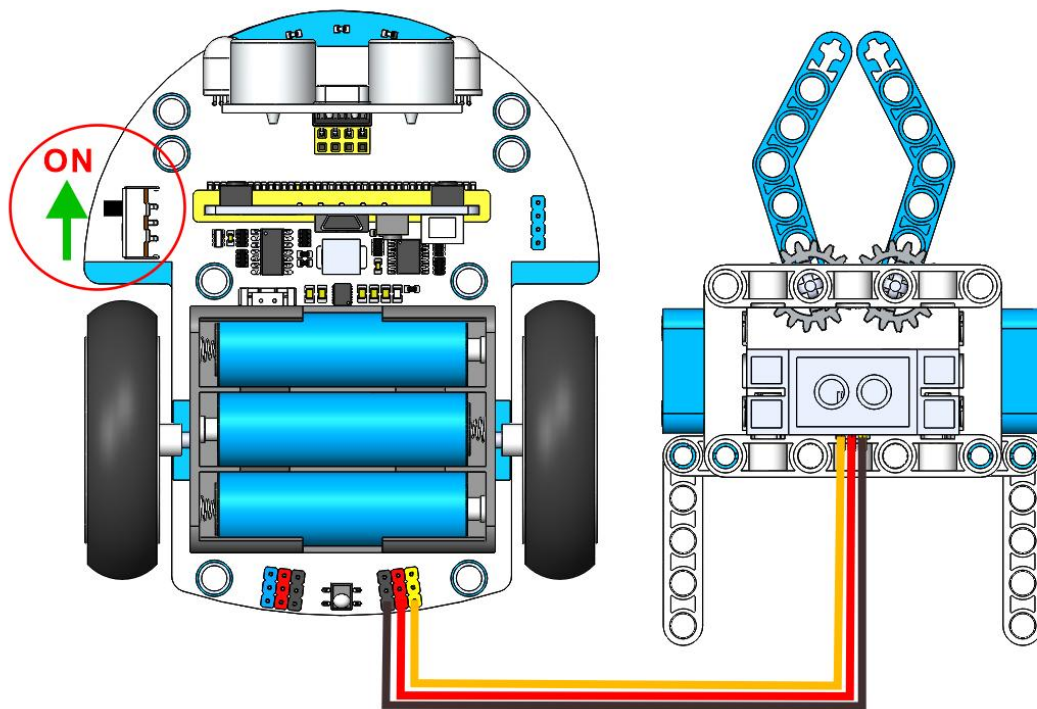
Wiring



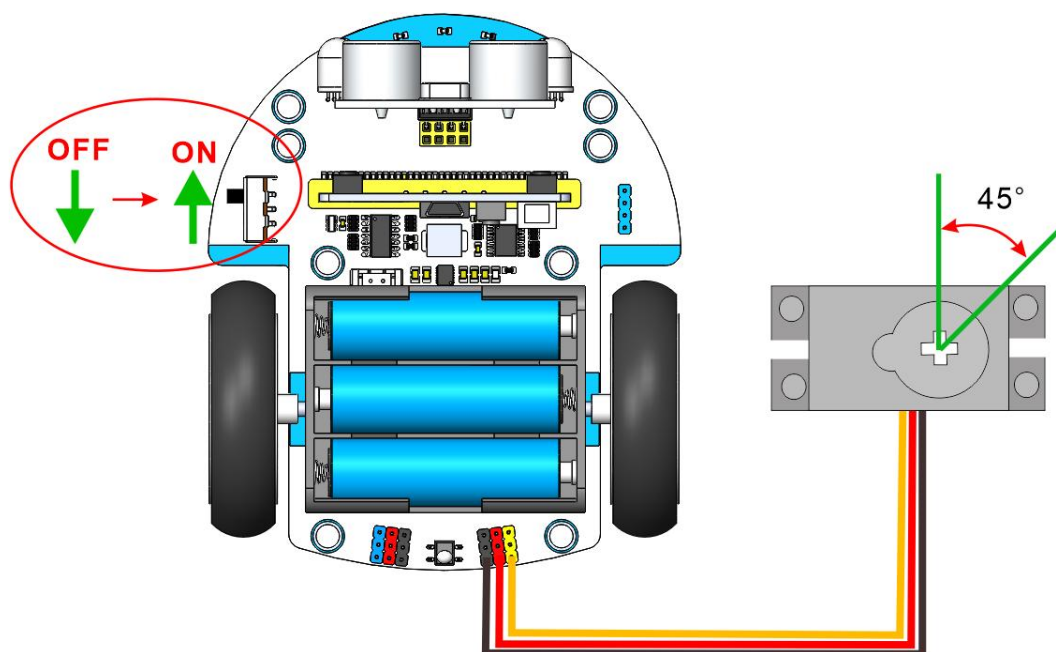
Upload code to Pybit



Power-up the pybit and initialize the servo

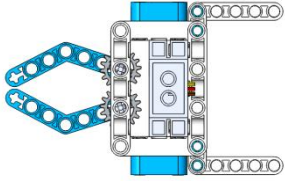


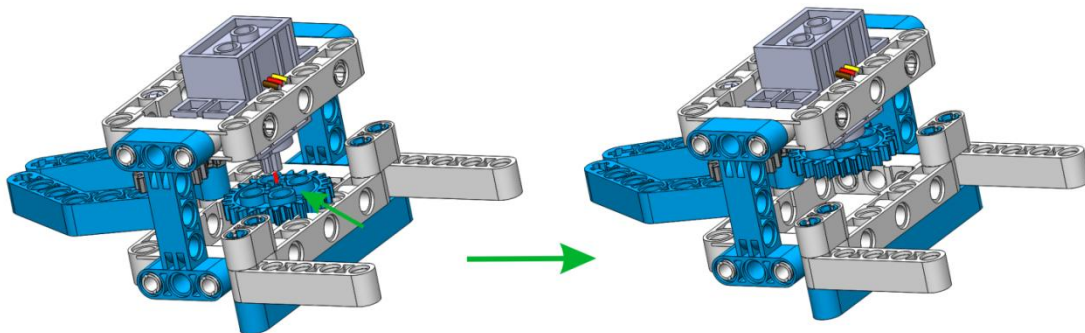
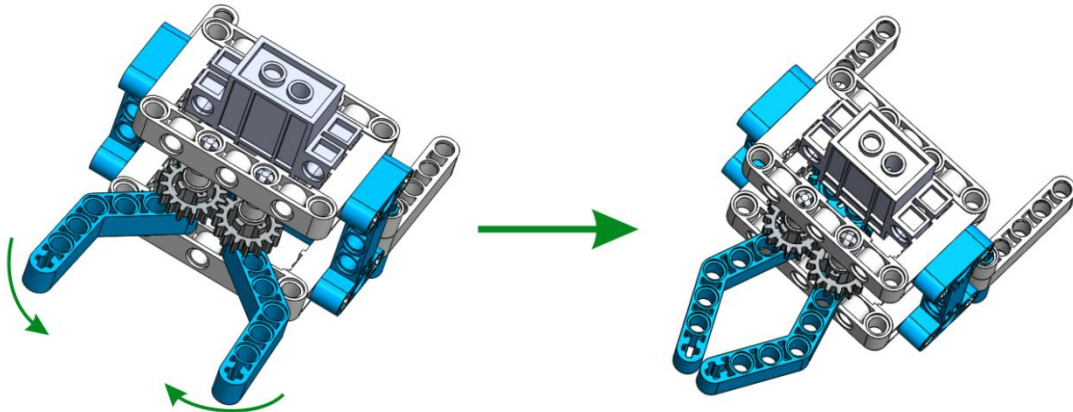
Check if the servo has been properly initialized



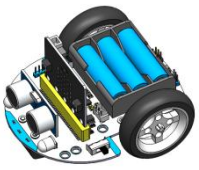
If the servo rotates by any 45-degree angle after each power-off and restart, it indicates successful initialization of the servo angle.

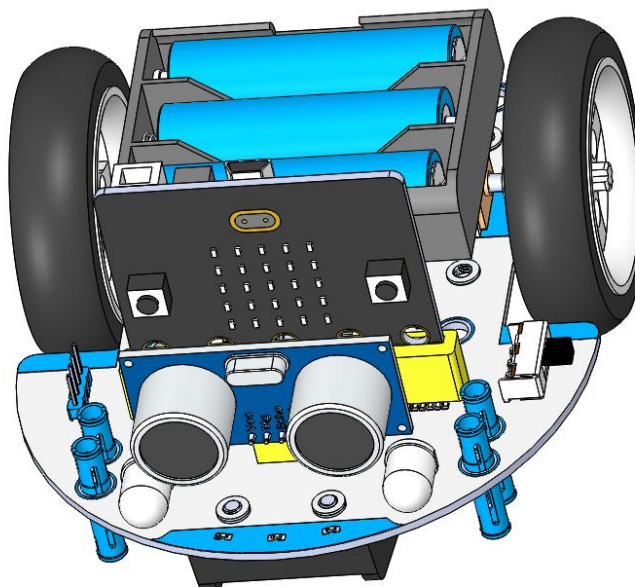
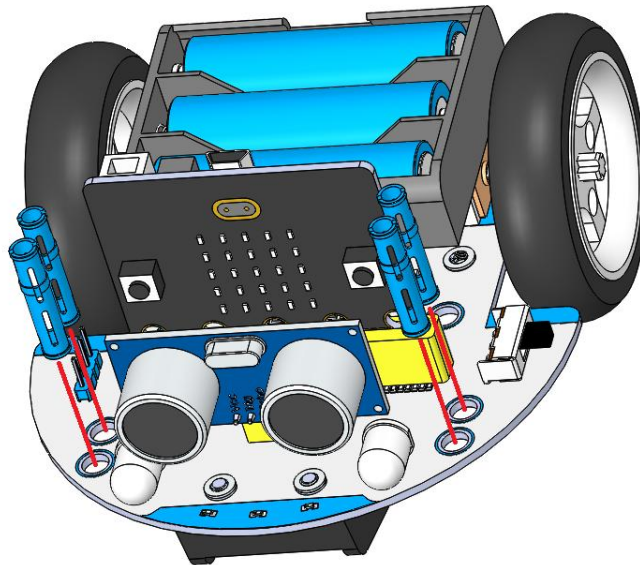
Step 19: Attach the servo gear

Component from Step 18	3648 24-Tooth Gear 1PCS
	

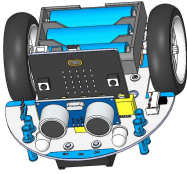


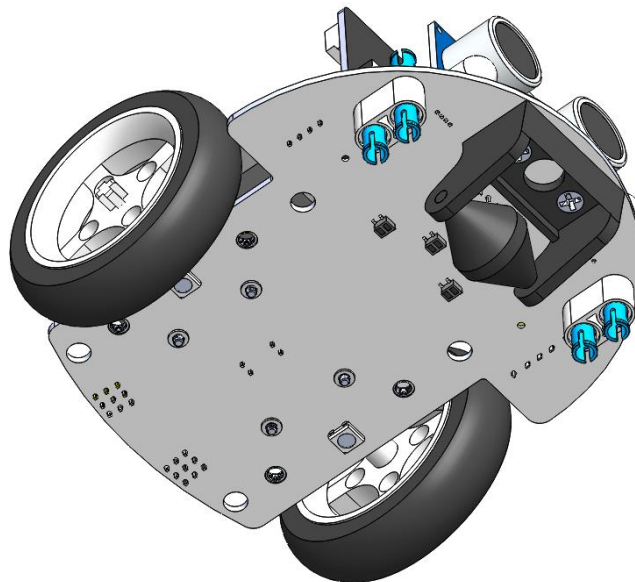
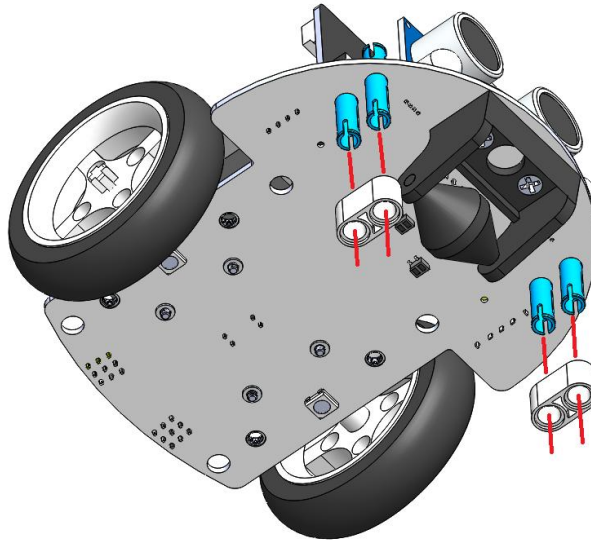
Step 20: Mount pins onto the Pybit

Pybit car with micro:bit v2	6558 Friction Pin 4PCS
	

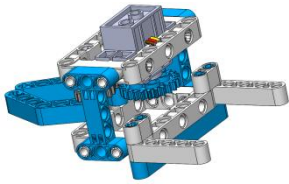
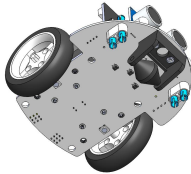


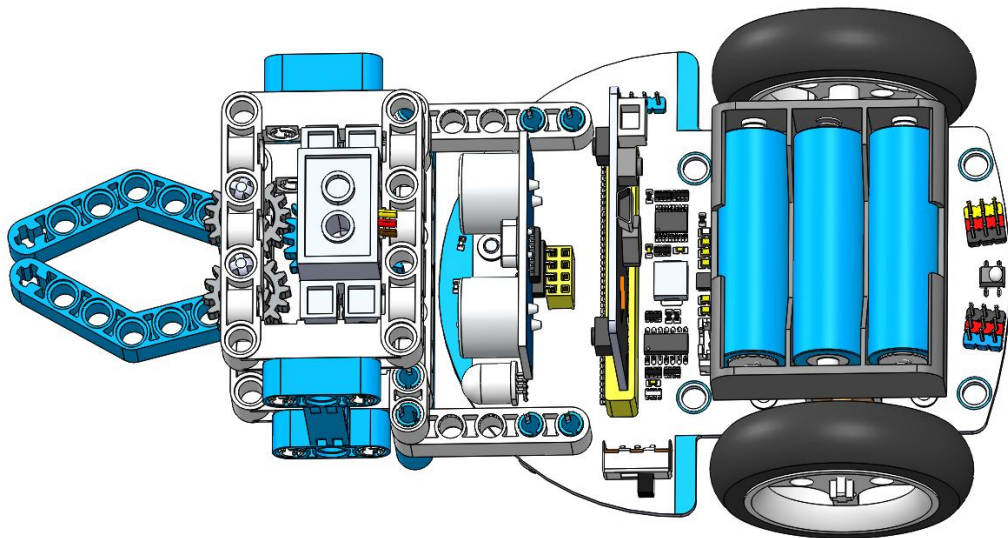
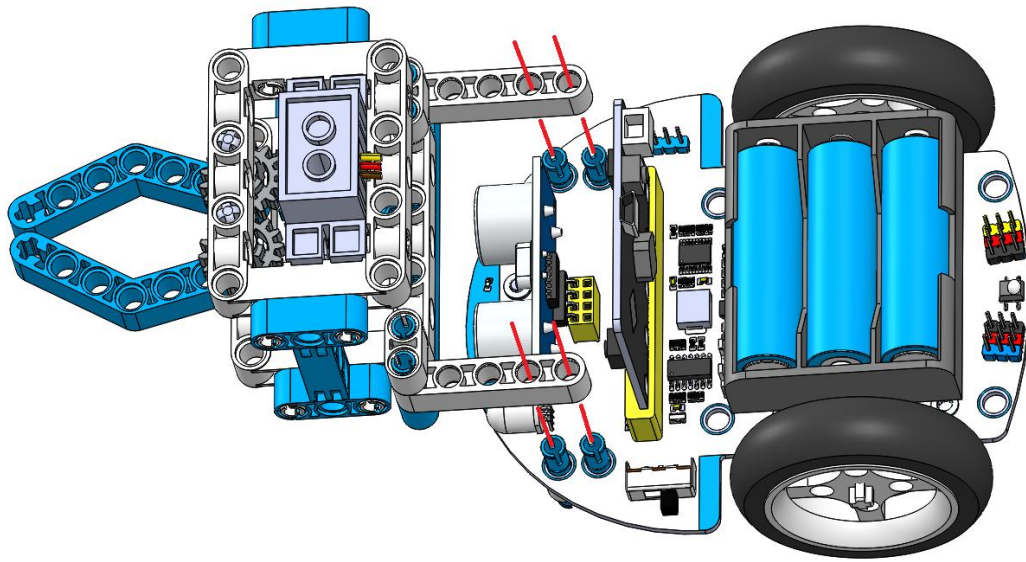
Step 21: Secure the pins on the Pybit

Component from Step 20	43857 2-Hole Beam 2PCS
	



Step 22: Attach the gripper to the Pybit

Component from Step 19	Component from Step 21
	



5.4 MakeCode Editor

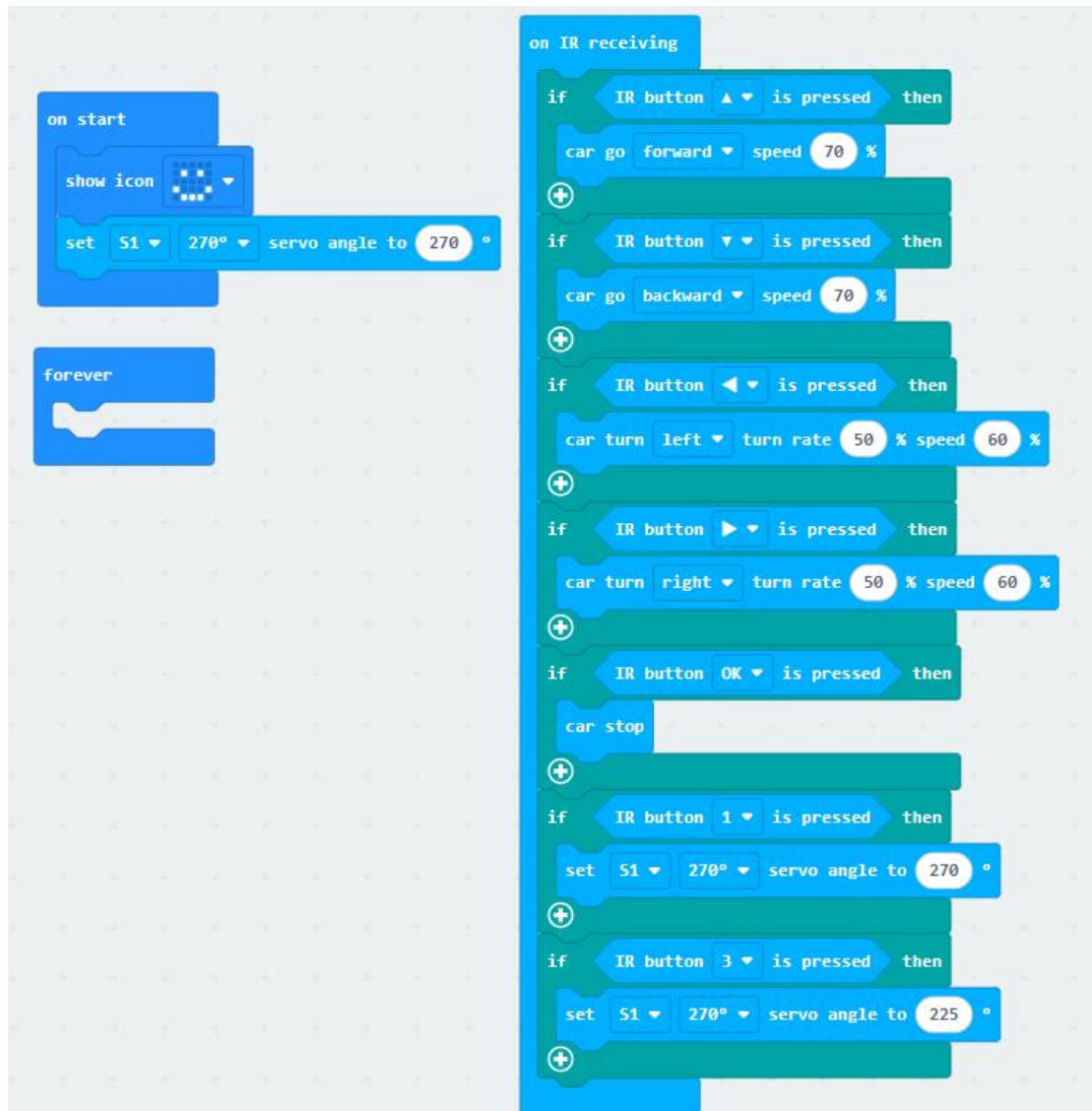
Visit <https://makecode.microbit.org/>

5.5 Example Code

You can directly upload the code we provided to the Micro:bit on the Pybit car to control the competition car. The example code for the competition car is stored in the following path of the tutorial package: xx/xx/Code.

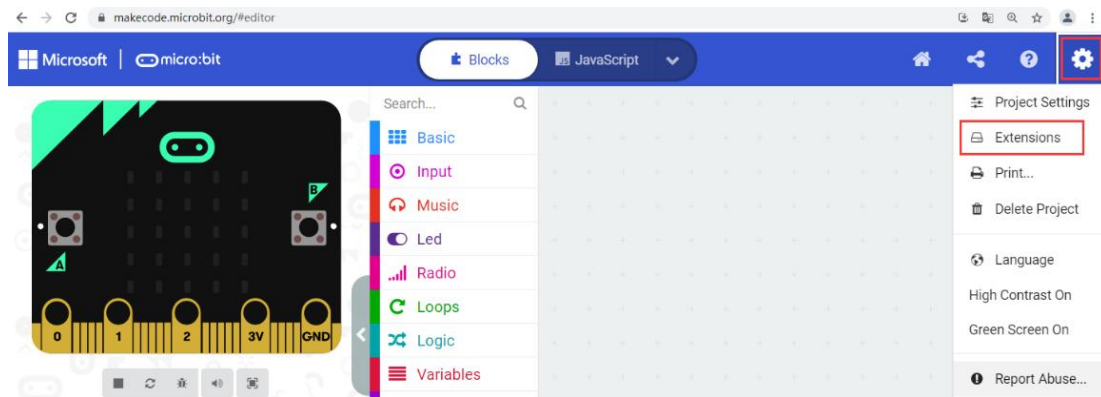


microbit-Clamp
-car.hex



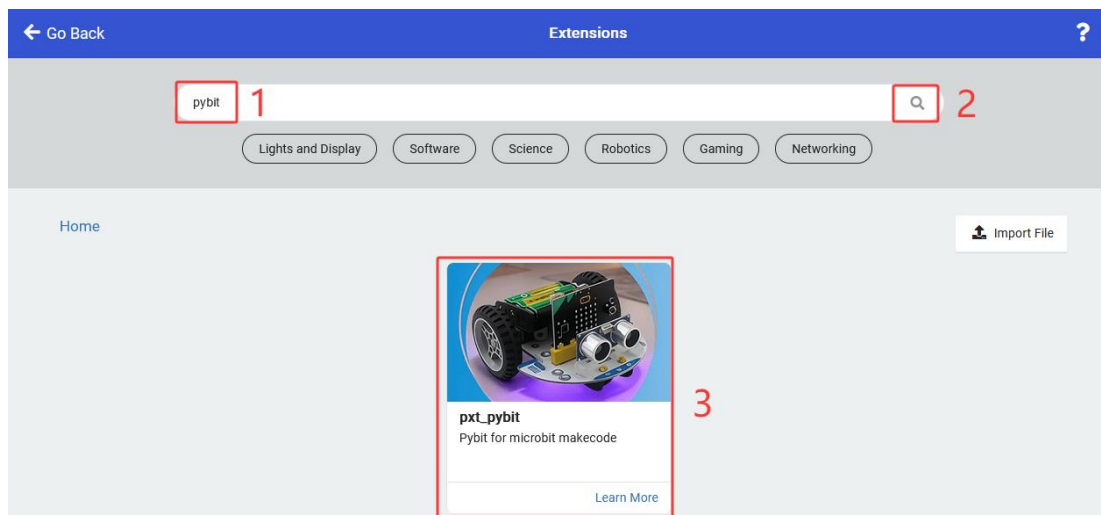
5.6 Extension Library

If you prefer to drag and drop code blocks to create the sample program yourself, you will need to add an extension library specifically developed for Pybit.

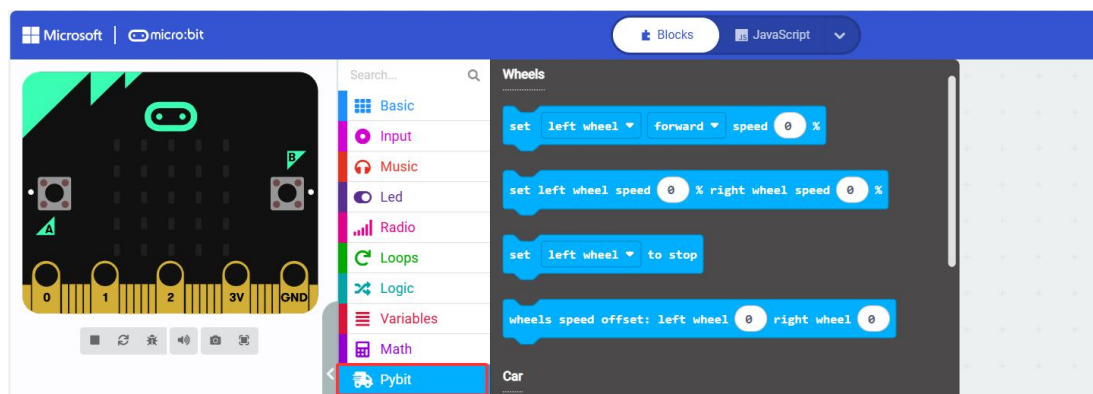


Copy "pybit" into the search box on the extensions page, then click the search button on the right.

Click the extension named **Pybit** from the search results.



After a few seconds, the page will redirect to the main MakeCode interface, where you will see the newly added Pybit extension in the toolbox list.



5.7 Result



: Press this button once to close the gripper



: Press this button once to open the gripper



: Press this button once – the car moves forward



: Press this button once – the car moves backward



: Press this button once – the car turns left



: Press this button once – the car turns right



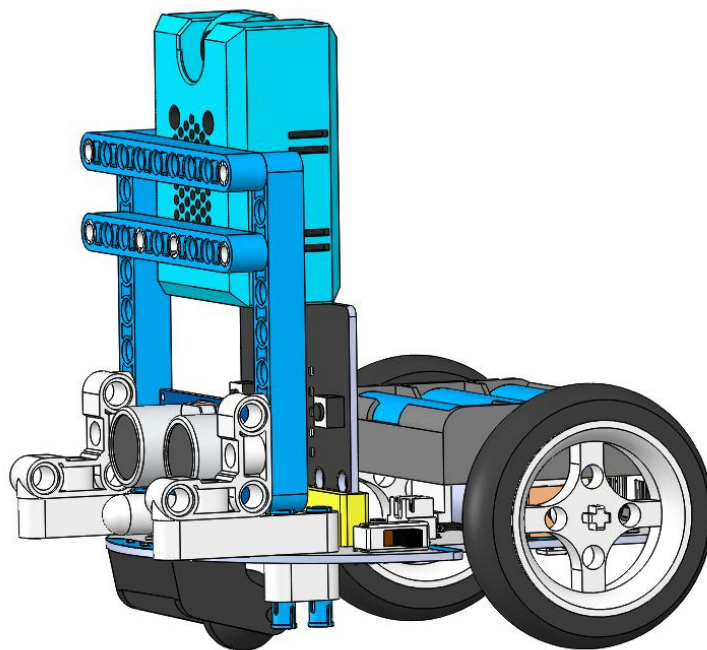
: Press this button once – the car stops.

6

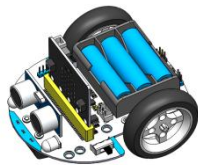
AI Road Sign Recognition Car

6.1 Goal

Build a Pybit car equipped with an AI camera that can recognize road signs and move according to their directional instructions.



6.2 Component Required

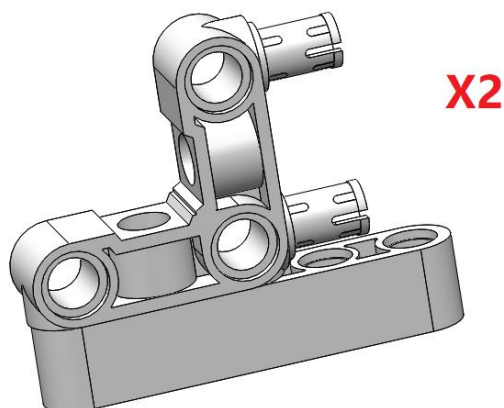
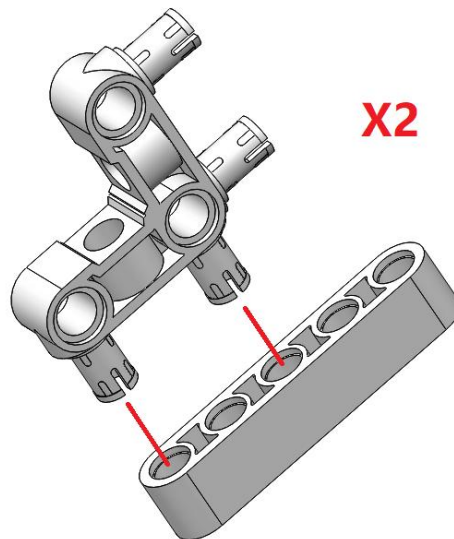
Pybit car with micro:bit v2	KOI2 AI camera 1PCS
	
Pybit expansion pack	
	

6.3 Assembly Steps

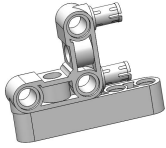
Construct an AI camera mount using standard building blocks, then install it on the Pybit.

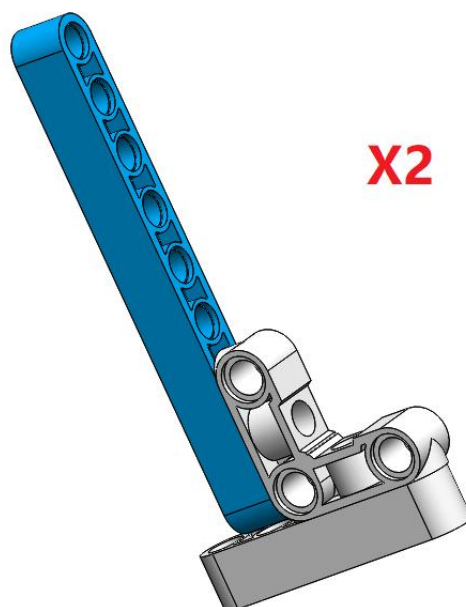
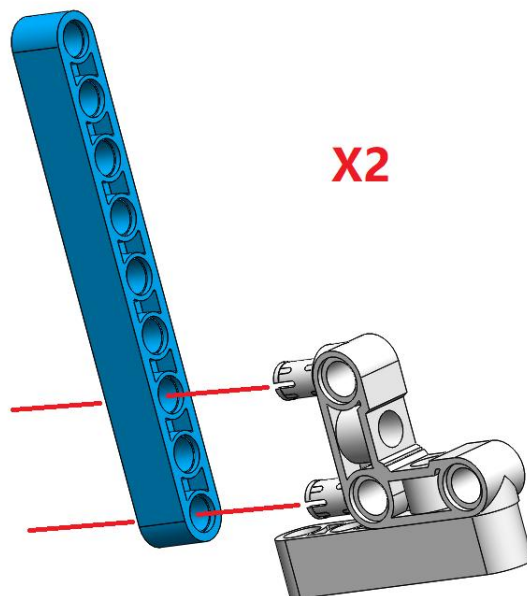
Step 1: Assemble the base

55615 Right-Angle Pin 2PCS	32316 5-Hole Beam 2PCS
	



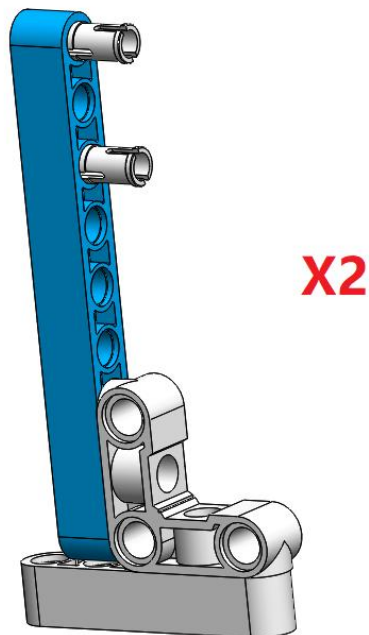
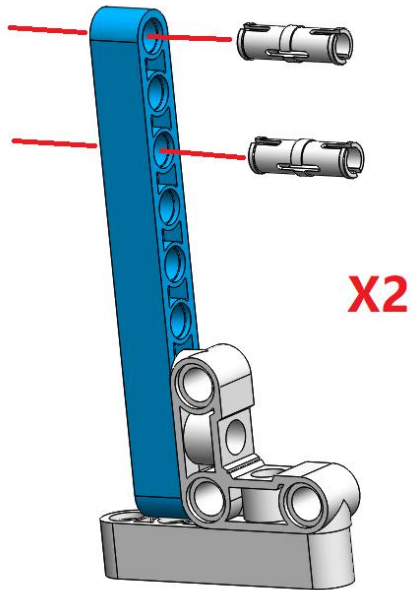
Step 2: Construct the support pillar

Component from Step 1 2PCS	40490 九孔臂梁 2PCS
	



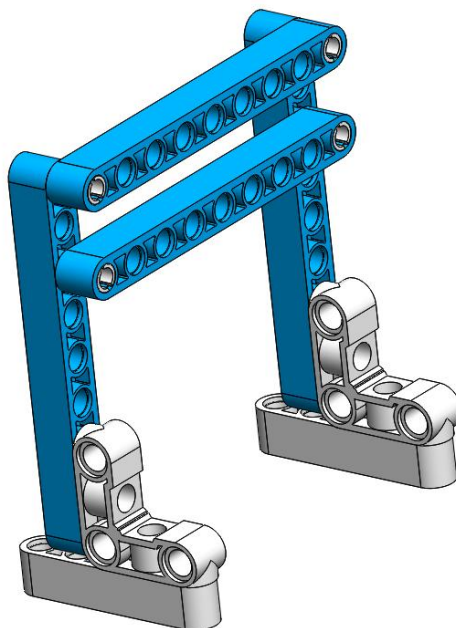
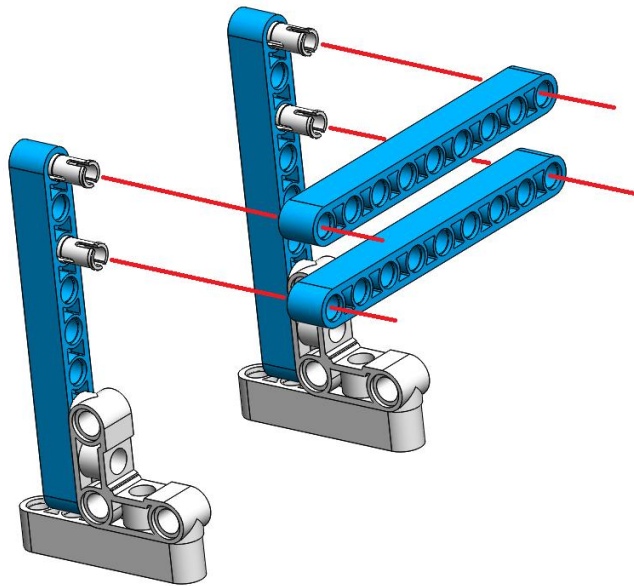
Step 3: Install the crossbeam pins

Component from Step 2 2PCS	2780 Friction Pin 4PCS
	

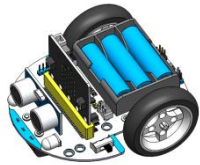


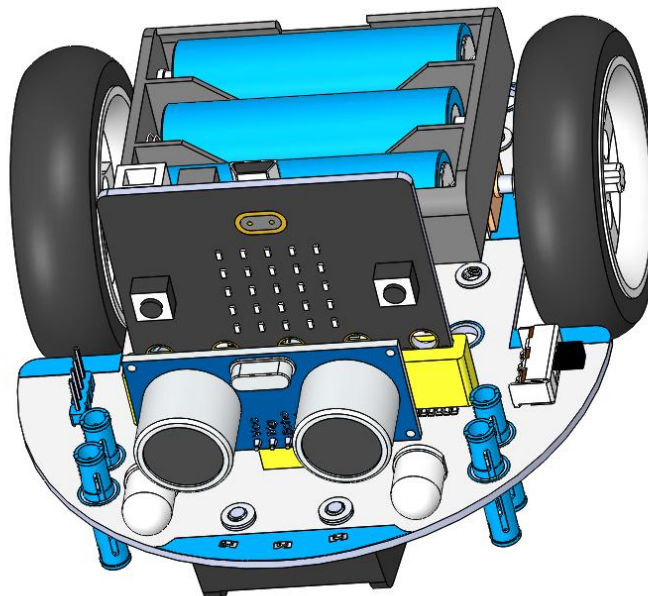
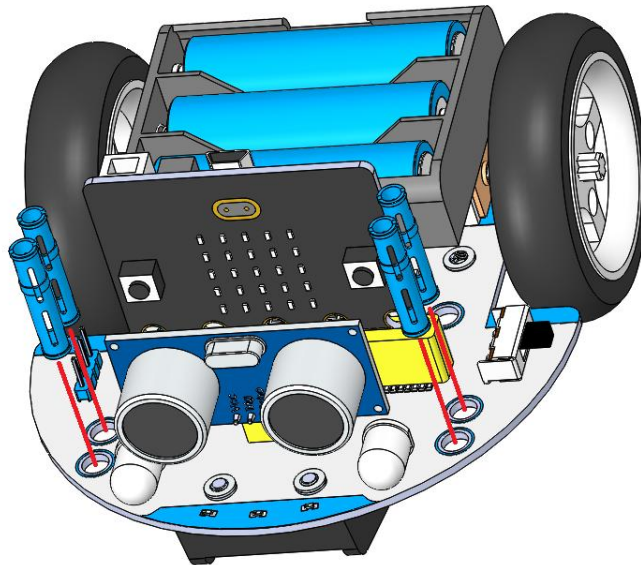
Step 4: Assemble the crossbeam

Component from Step 3 2PCS	40490 9-Hole Beam 2PCS
	

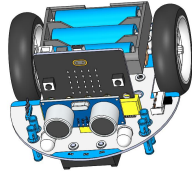


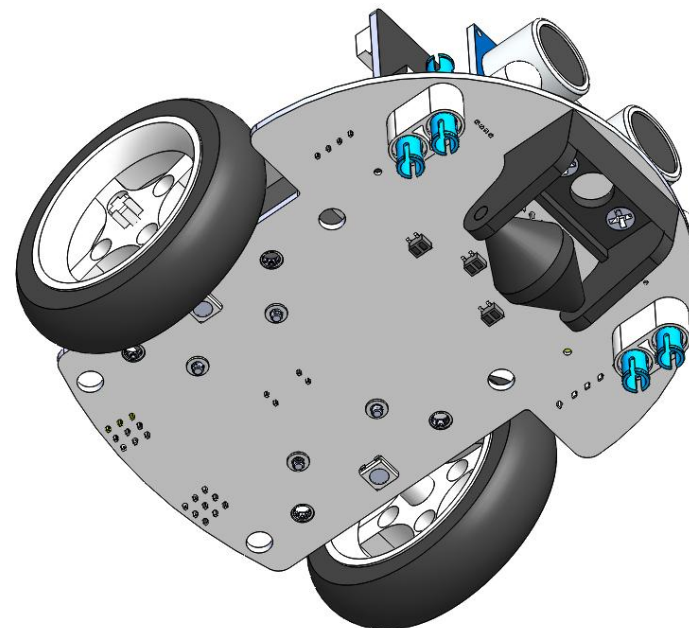
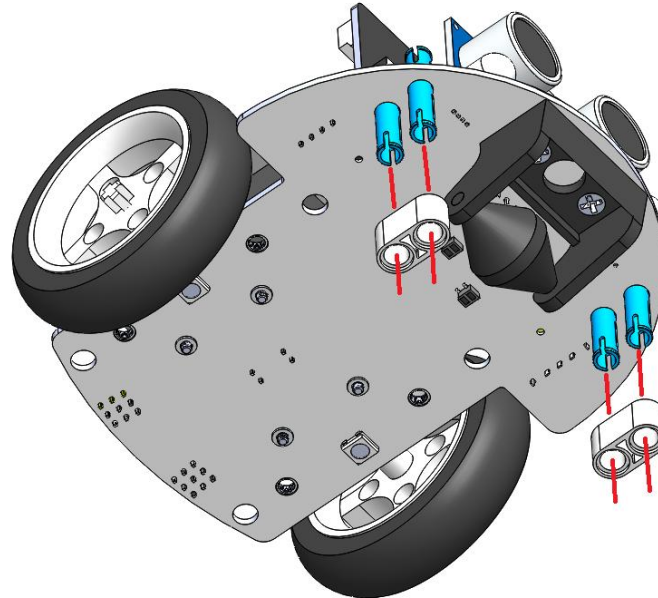
Step 5: Mount the pins onto the Pybit

Pybit car with micro:bit v2	6558 Friction Pin 4PCS
	

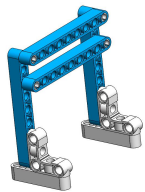
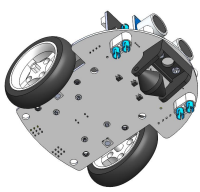


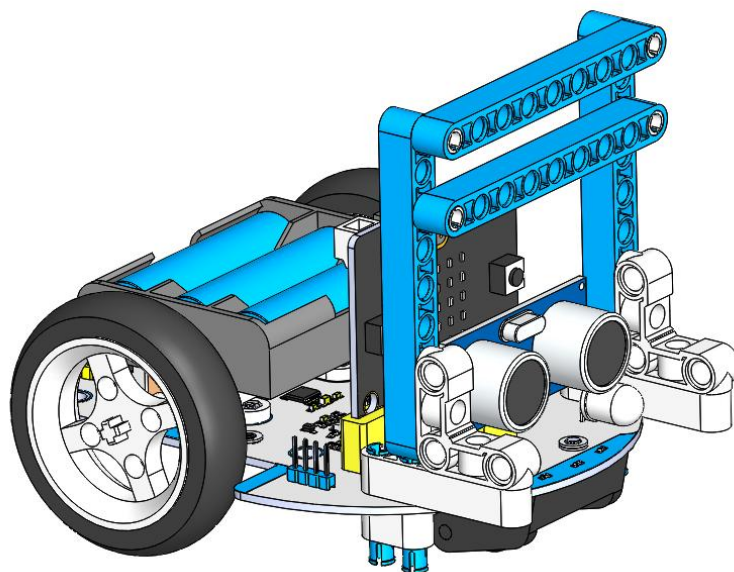
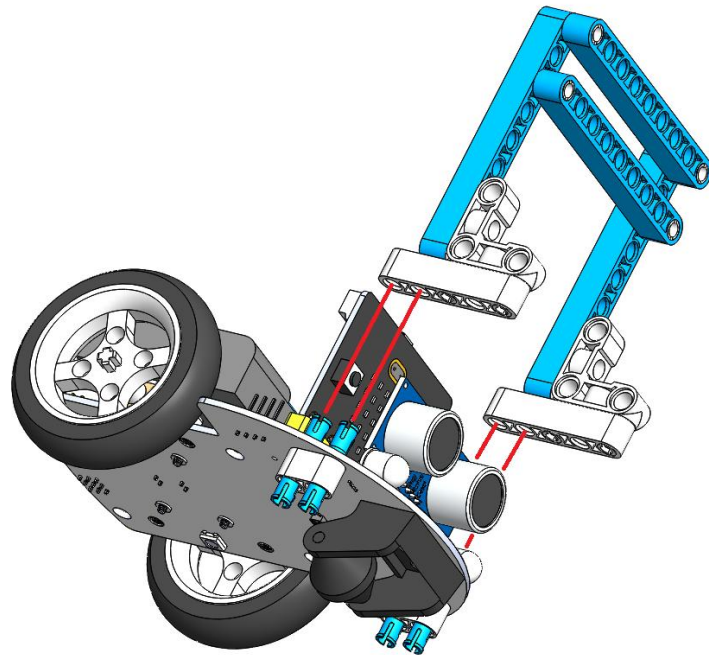
Step 6: Secure the pins on the Pybit

Component from Step 5	43857 2-Hole Beam 2PCS
	

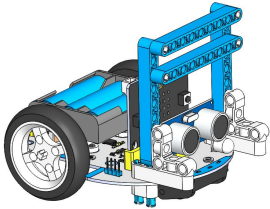


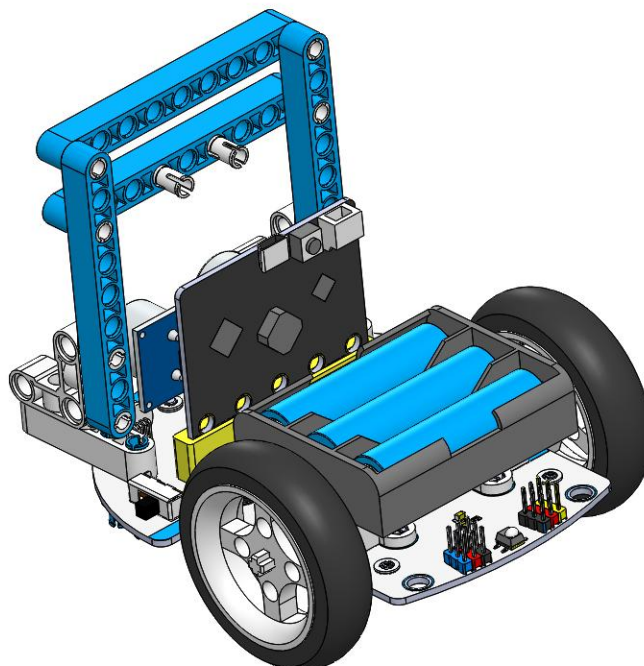
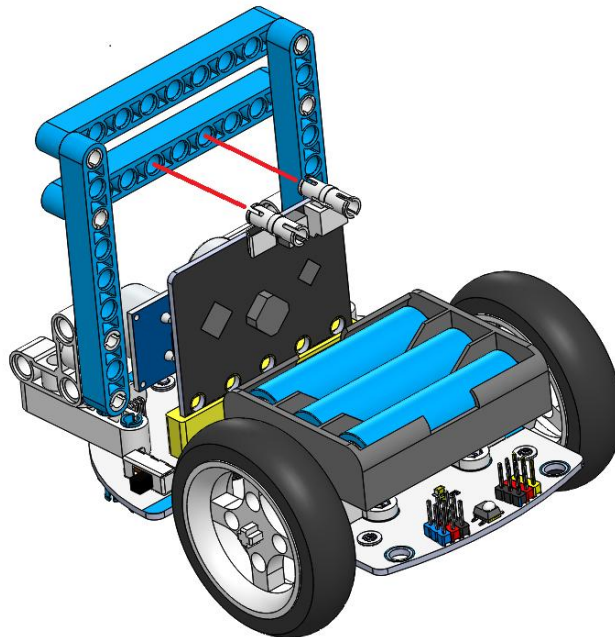
Step 7: Attach the bracket to the Pybit

Component from Step 4	Component from Step 6
	

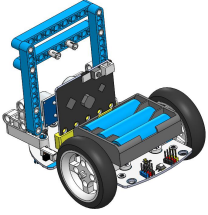


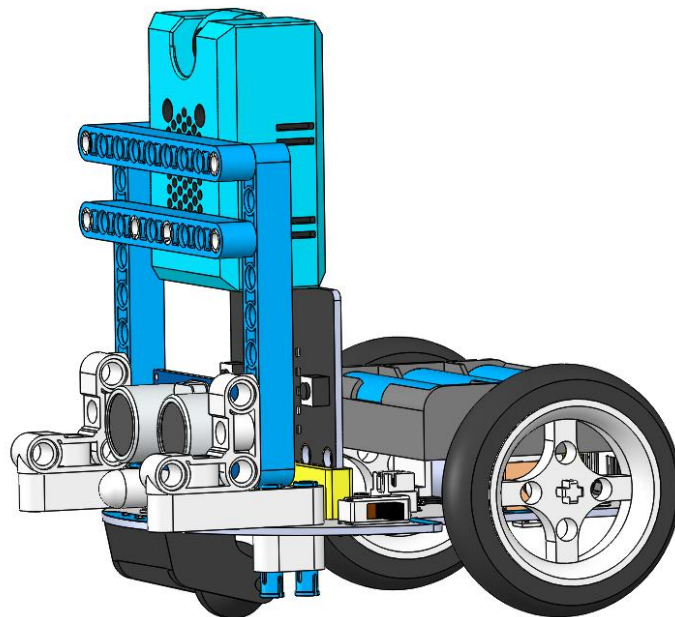
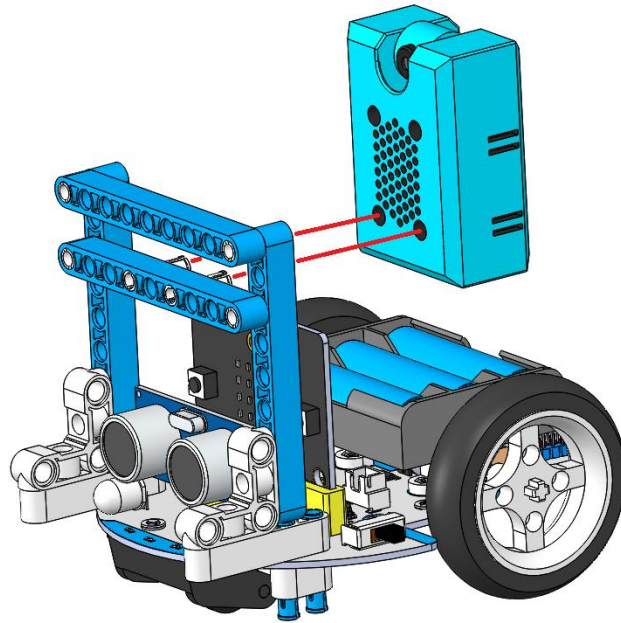
Step 8: Install the pins for AI camera

Component from Step 7	2780 Friction Pin 2PCS
	



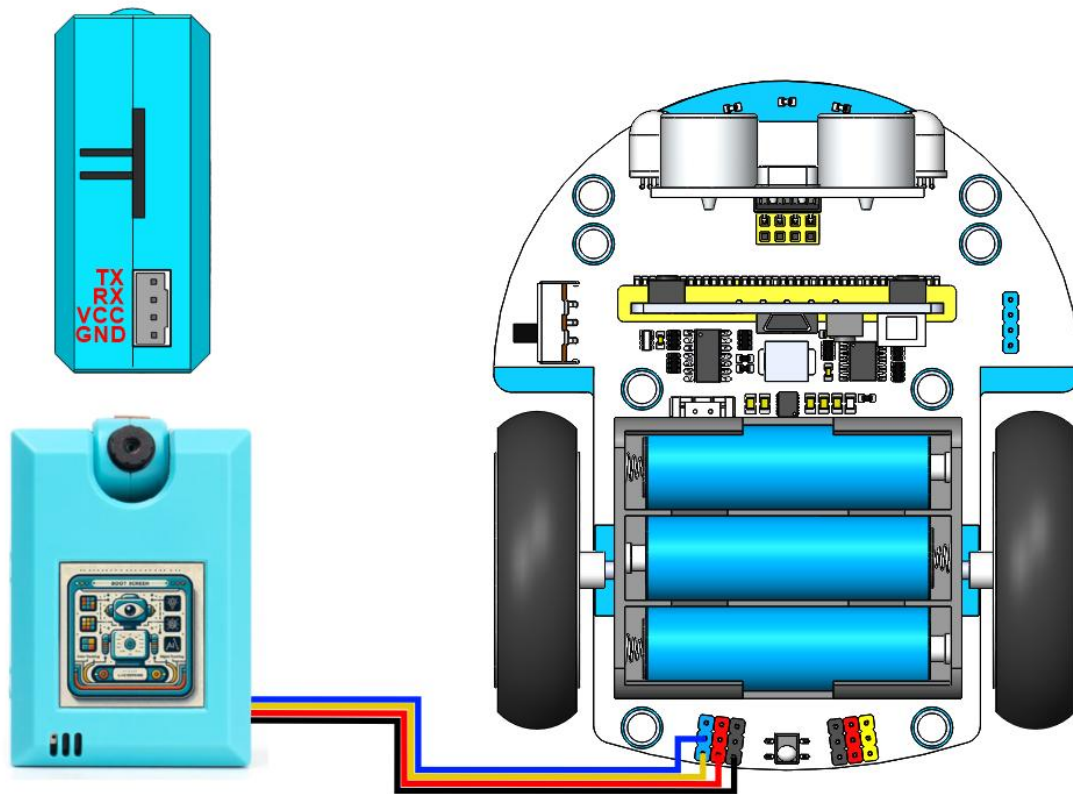
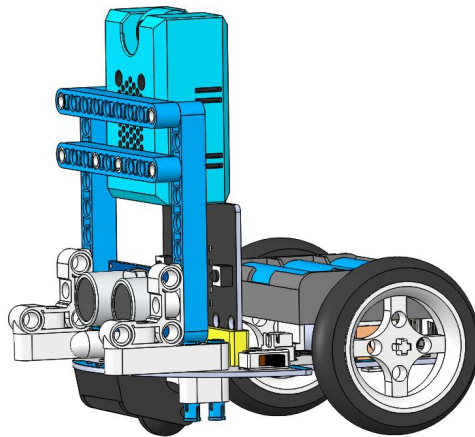
Step 9: Mount the AI camera to the Pybit

Component from Step 8	KOI2 AI Camera 1PCS
	



Step 10: Wiring

Component from Step 9



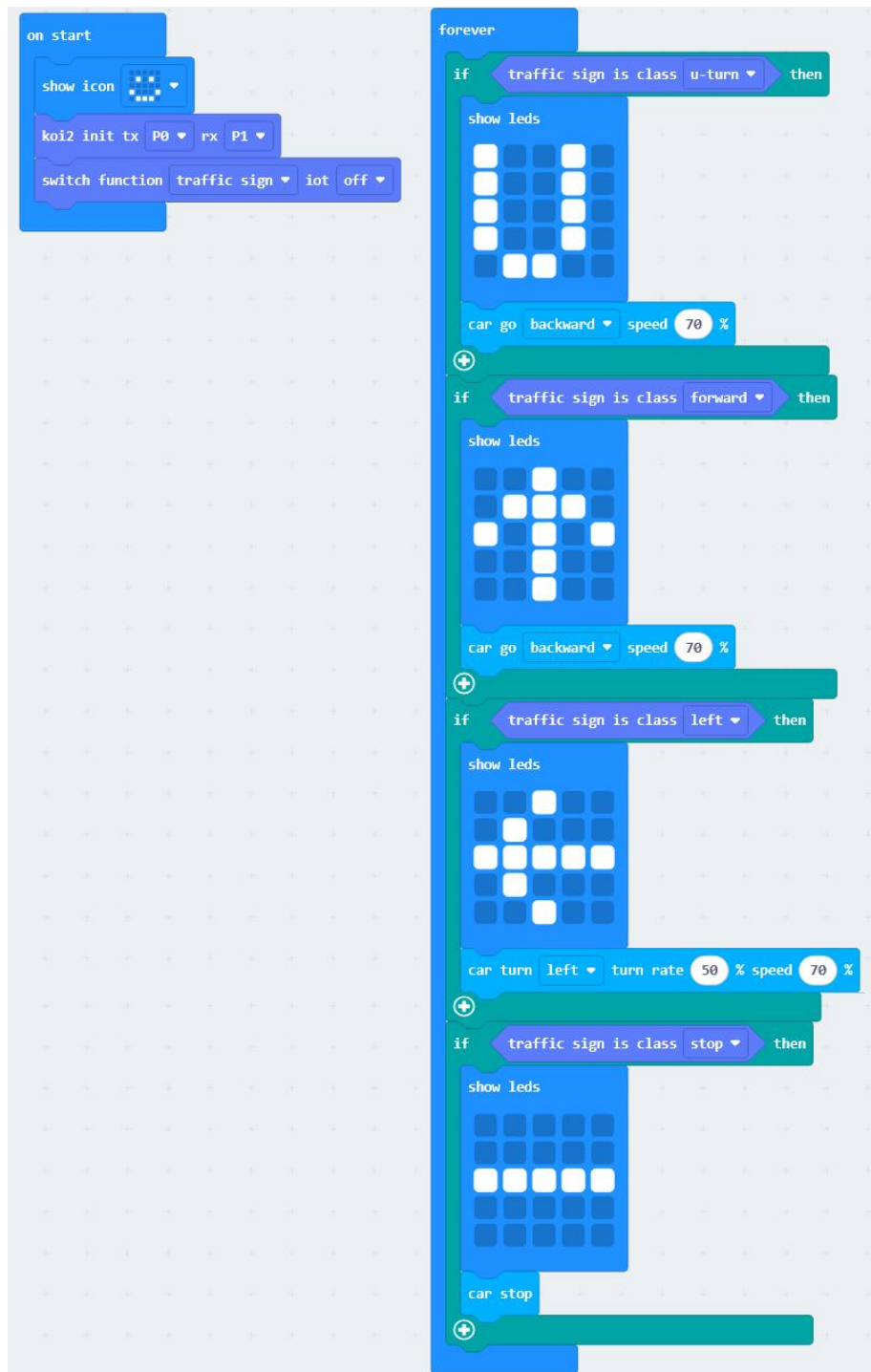
Pybit	P1	P0	3V3	G
KOI2	TX	RX	VCC	GND

6.4 MakeCode Editor

Vist <https://makecode.microbit.org/>

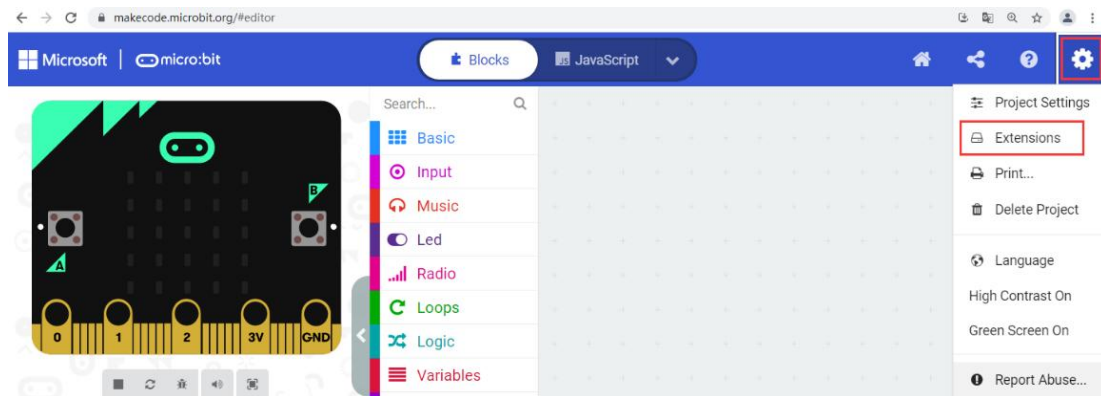
6.5 Example Code

You can directly upload the code we provided to the Micro:bit on the Pybit car to control the competition car. The example code for the competition car is stored in the following path of the tutorial package: xx/xx/Code.



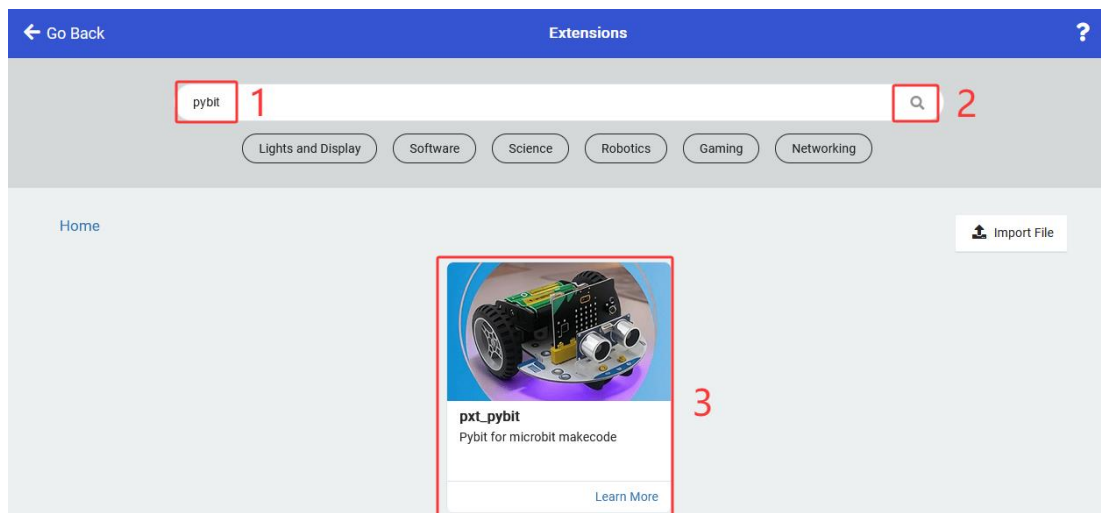
6.6 Extension Libraries

If you prefer to drag and drop code blocks to create the sample program yourself, you will need to add an extension library specifically developed for Pybit.

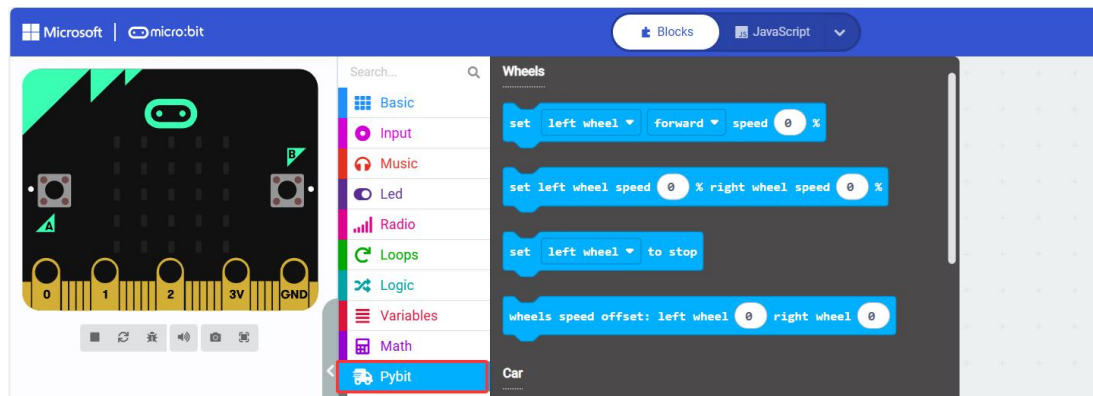


Copy "**pybit**" into the search box on the extensions page, then click the search button on the right.

Click the extension named **Pybit** from the search results.

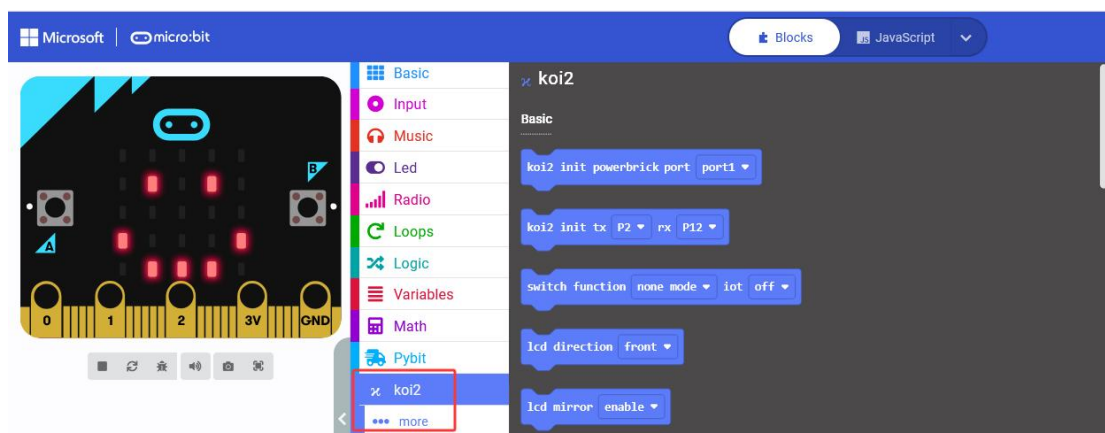
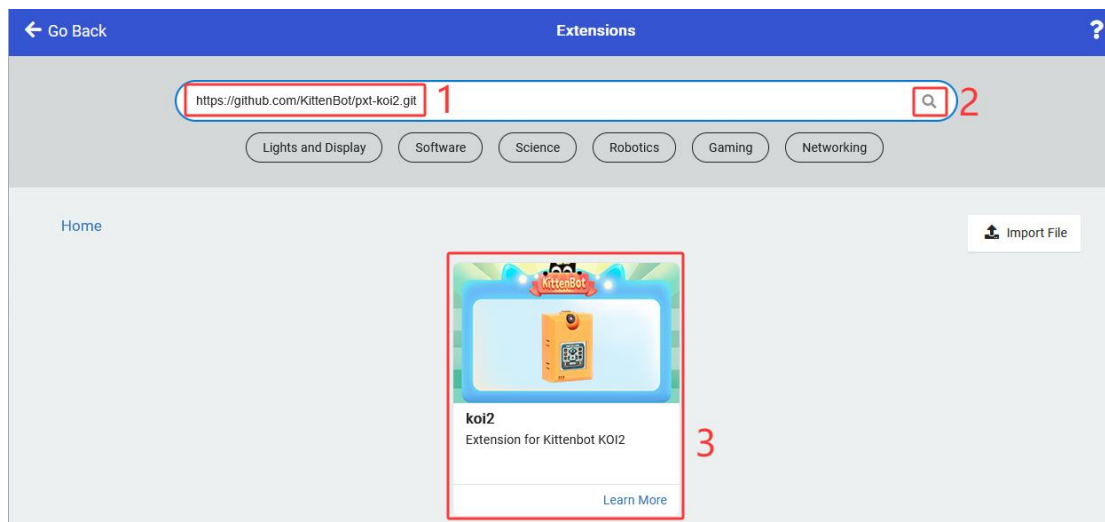


After a few seconds, the page will redirect to the main MakeCode interface, where you will see the newly added Pybit extension in the toolbox list.



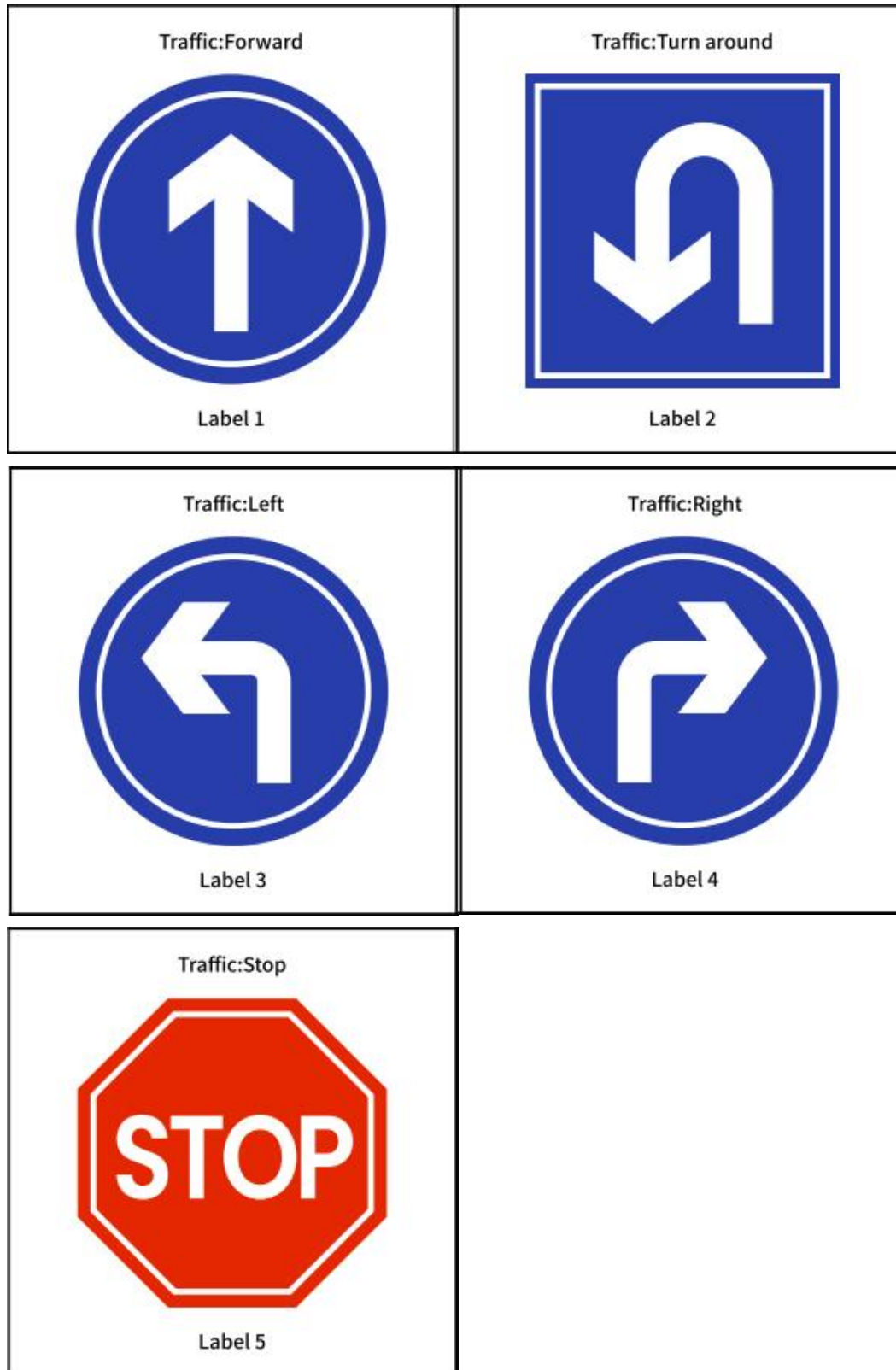
Add KOI2 extension

URL: <https://github.com/KittenBot/pxt-koi2.git>



6.7 Result

The camera will recognize the following road sign icons and move according to the directions indicated by the icons:



6.8 Other Recommended AI Cameras

Pybit is also compatible with AI cameras from other brands. Their purchase links are listed below:

ELECTFREAKS AI camera	Sentry2 AI camera	Sengo2 AI camera
		

The official websites of these brands provide detailed tutorials for their respective AI cameras. You can refer to these tutorials to program even more interesting functions!

7

Contact us

To help us assist you quickly, please have the following information ready:

Your order number.

Product model (e.g., M1C0001) and software version (e.g., MakeCode v3.0.XX).

A detailed description of the issue or your question.

Steps you have already tried.

Any relevant error message screenshots, photos, or code snippets.

Other Inquiries?

We value your input and are always looking to improve. Please feel free to reach out for:

Tutorial Errors & Feedback: Help us make our documentation better.

Product Ideas & Suggestions: We'd love to hear your great ideas.

Partnerships & Collaboration: Interested in working together? Let's talk.

Discounts & Promotions: Inquire about educational or bulk pricing.

Anything Else: For all other non-technical questions.

Support Channels



support@siyeenove.com



<https://siyeenove.com>